

Cross-species insemination reveals mouse sperm ability to enter and cross the fish micropyle

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eLife Assessment

This **important** study reports the conservation of sperm-egg envelope binding by demonstrating successful recognition of the micropyle in fish eggs by mouse sperm. The evidence supporting the conclusions drawn is **convincing**. This study will be of interest to reproductive biologists and clinicians studying the biology of fertilization and fertility.

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Abstract

Extracellular matrices surrounding eggs in fish (chorion) and mammals (zona pellucida, ZP) regulate gamete recognition before fertilization, though their mechanisms differ. Mouse sperm bind and cross the ZP at any site, while fish sperm cross the chorion through a funnel-shaped opening, the micropyle. To explore these divergent processes, we established cross-species insemination assays, mixing zebrafish eggs with mouse sperm. While mouse sperm could not bind to the chorion, a subpopulation successfully located and crossed the fish micropyle. Confocal and electron microscopy revealed that sperm entered the micropyle and accumulated in the zebrafish inter-chorion space. However, transgenic mouse sperm with mCherry-labeled acrosomes failed to undergo acrosome exocytosis efficiently in the micropyle, with both acrosome-intact and reacted sperm found in the inter-chorion space. Sperm entry and crossing were dependent on hyperactive motility, as sperm from *CatSper1*^{Null} mice, which fail to undergo hyperactivation, did not interact with or cross the micropyle. These findings suggest a conserved mechanism for sperm entry into the micropyle, providing a novel platform to investigate cross-species gamete interactions and uncover novel steps in fertilization.

Impact Statement

This study reveals that while mouse sperm cannot bind to the zebrafish chorion, a subset of hyperactivated sperm can enter through the micropyle, suggesting a conserved mechanism of gamete interaction across species.

Introduction

Studies of interspecific gamete interaction have been historically important for elucidating the mechanisms regulating fertilization (VACQUIER, 1979 [↗](#); Vieira & Miller, 2006 [↗](#); Yanagimachi et al., 1976 [↗](#)), which is required for the successful development of fish and mammals (Bhakta et al., 2019 [↗](#); Deneke & Pauli, 2021 [↗](#)). Despite decades of investigation, however, the understanding of the molecular basis of sperm-egg interaction remains incomplete.

In mice, upon mating, hundreds to thousands of sperm leave the uterus and cross the uterotubal junctions, which separate the uterus from the oviduct (Avella et al., 2016 [↗](#); Ded et al., 2020 [↗](#); Suarez, 2016 [↗](#)). These sperm bind to the epithelial cells of the isthmus in the oviduct until eggs are released from the ovary during ovulation into the ampulla of the oviduct (Hunter, 1993 [↗](#)), where fertilization occurs. A portion of these sperm undergoes hyperactivation, which consists of a sudden change in the motility pattern of the sperm flagella, generating an asymmetrical beat that mediates the sperm release from the isthmus (Han et al., 2016 [↗](#)). Hyperactive sperm enter the ampulla of the oviduct and traverse the cumulus mass, a hyaluronic-interspersed mass of cells, remnant of the granulosa cells from the follicle. Each cumulus mass in the ampulla may enclose a single Metaphase II (MII) egg, mature and competent for fertilization. Upon reaching the egg, sperm bind to the zona pellucida, an extracellular glycoprotein matrix composed of three glycoproteins defined as ZP1, ZP2, and ZP3. After binding, the fertilizing sperm cross the zona and fuse with the oolemma (Bhakta et al., 2019 [↗](#)).

In fish, the sperm do not bind to the chorion surrounding the ovulated egg (Mold et al., 2001 [↗](#)). Instead, sperm cross the chorion through a narrow, funnel-shaped canal defined as the micropyle (Yi et al., 2019 [↗](#)). Early studies in the Pacific herring (*Clupea sp.*) show that, once released into seawater, sperm are virtually motionless until they come near the micropyle area of an egg, whereupon they become motile (Yanagimachi, 1957 [↗](#); Yanagimachi et al., 2017 [↗](#)). In bitterling, zebrafish, and other fish species, sperm contact with water is sufficient to activate their motility, yet, as they pass near the micropyle area, their motility increases (Suzuki, 1958 [↗](#)).

Because fish sperm bypass binding to the chorion using the micropyle, fish zona proteins have always been assumed to lack the ability to support sperm binding (Mold et al., 2001 [↗](#)). Zebrafish chorion is composed of two proteins, ZP2 and ZP3, homologs to the mammalian zona proteins (Mold et al., 2001 [↗](#)). Like other external fertilizing species, zebrafish sperm cross the chorion through the micropyle before fusing with the egg. On the other hand, mammalian eggs do not present a micropyle in the zona pellucida. Hence, sperm can cross at any binding site of the zona matrix, as shown by the presence of vesiculated acrosomal shrouds in multiple sites of the zona surface (VandeVoort et al., 1997 [↗](#); Wakayama et al., 1996 [↗](#); Yanagimachi & Phillips, 1984 [↗](#)).

The abovementioned observations raise the prediction that mouse sperm may not recognize zebrafish ZP2 or ZP3 and would therefore be unable to bind to the zebrafish chorion. Additionally, mouse sperm may not recognize the micropyle, given that no such structure exists in mammalian zonae pellucidae. Importantly, these two predictions have never been experimentally tested. As testing these predictions could provide valuable insights into the molecular basis of species

specificity in gamete attraction and recognition prior to fertilization, the overall aim of this study was to determine whether mammalian sperm can recognize and respond to zona proteins or the micropyle in zebrafish eggs.

To test these predictions, we have established a mouse-fish gamete insemination assay that consists of mixing fish eggs with mouse sperm. We show that mouse sperm could not bind to individual zebrafish ZP2/ZP3 nor to the chorion. Yet, unexpectedly, a subpopulation of mouse sperm recognized and entered the micropyle and accumulated in the oocyte inter-chorion space (ICS; the space between the inner aspects of the chorion and the oocyte). To capitalize on such discoveries, we used mouse genetics to show that only hyperactive mouse sperm could enter and cross the micropyle. We also documented that the zebrafish micropyle could not effectively induce acrosome exocytosis. Our studies provide a novel platform for cross-species insemination experiments and have implications for the mechanisms by which sperm can locate and enter micropyle of the egg.

Results

Zebrafish zona pellucida proteins are not sufficient to support mammalian sperm binding

Mouse zona pellucida is composed of three glycoproteins: ZP1, ZP2, and ZP3, and the N-terminus of ZP2 has been identified as the ligand necessary and sufficient to support mouse sperm binding (Avella et al., 2014 [↗](#); Tokuhiro & Dean, 2018 [↗](#)). In contrast, the zebrafish chorion contains only two glycoproteins, ZP2 and ZP3 (Mold et al., 2001 [↗](#)), and zebrafish sperm do not rely on these proteins for interaction with the chorion. Therefore, while zebrafish ZP2 and ZP3, either individually or in combination, are not predicted to mediate sperm interaction, this has not been experimentally tested and remains unknown. To assess this, we used mouse sperm, which are known to require the N-terminus of ZP2 for binding to the zona pellucida (Avella et al., 2014 [↗](#); Tokuhiro & Dean, 2018 [↗](#)), as a functional assay to test the ability of zebrafish ZP2 and ZP3 to support sperm binding.

Zebrafish ZP2 isoforms present 31.09-33.09% identity to mouse ZP2 (Figure 1A [↗](#)). The first 6-cysteine residues of zebrafish ZP2 are organized into a trefoil domain, and zebrafish ZP2 lacks the homologous domain to the mammalian N-terminal region (Figure 1A [↗](#)), which has been reported to be necessary and sufficient to support mouse and human sperm binding (Avella et al., 2014 [↗](#); Tokuhiro & Dean, 2018 [↗](#)). Recombinant baculovirus encoding zebrafish ZP2 and ZP3 were generated (Figure 1B [↗](#)). Each recombinant protein presented at the N-terminus a gp67 signal peptide from *Autographa californica* nuclear polyhedrosis virus (AcMNPV) (38 aa) to ensure proper secretion, and at the C-terminus, a 6-His tag to enable purification. Zebrafish ZP2²⁵⁻⁴⁰⁵ and ZP3²²⁻³⁹⁶ start at the N-terminus of the secreted ectodomains and include an even number of cysteine residues (Figure 1A [↗](#)). Confocal images and quantification with boxplots showed that mouse sperm bound sparsely to beads coated with either zebrafish ZP2 or ZP3 and to beads coated with both zebrafish ZP2 and ZP3, yet bound efficiently to beads coated with mouse ZP2 (Figure 1C [↗](#) and D [↗](#)) (Avella et al., 2016 [↗](#)). Therefore, we could conclude that beads coated with individual zebrafish ZP2 or ZP3, or with both ZP2 and ZP3, cannot support sperm binding *in vitro* as efficiently as mouse ZP2 (Figure 1D [↗](#)).

Mouse sperm do not bind to the zebrafish chorion

To further validate these observations, we used native zebrafish chorion (Mold et al., 2001 [↗](#)). First, to test whether zebrafish chorion remains intact under mouse *in vitro* fertilization (IVF) conditions, zebrafish ovulated eggs (n = 100, 3 replicates) were incubated at 37 °C, 5% CO₂ in

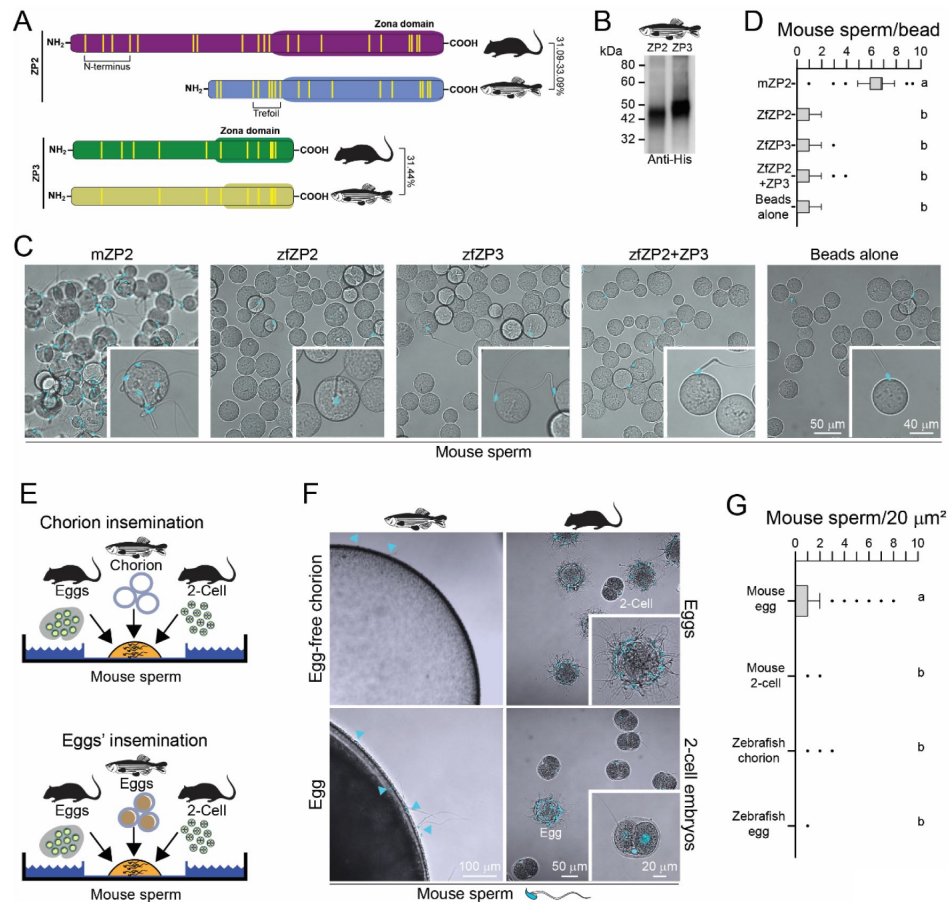


Figure 1.

Mouse sperm do not bind to zebrafish zona proteins.

A Schematic of zebrafish ZP2 and ZP3 peptides compared to mammalian homologs (Clustal Ω). Yellow bars represent cysteine residues. Percent values indicate identity between mouse and zebrafish amino acid sequences (NP_035905.1, mouse ZP2; AAK16578.1, zebrafish ZP2, variant A; AAK16577.1, zebrafish ZP2, variant B; AAK16579.1, zebrafish ZP2, variant C; NP_035906.1, mouse ZP3; NP_571406, zebrafish ZP3). **B** SDS-PAGE of recombinant zebrafish ZP2, ZP3, or ZP2 and ZP3 peptides (6-His mAb, immunoblot) expressed in Sf9 cells after purification from agarose beads. Molecular mass is indicated on the left. **C** Capacitated mouse sperm binding to beads carrying zebrafish (zf) ZP2, zfZP3 or zebrafish zfZP2 and zfZP3. Beads carrying mouse (m) ZP2 N-termini and beads alone were used as positive and negative controls, respectively. Differential Interference Contrast (DIC) (top) and confocal z projection (bottom) images, sperm nuclei (blue) stained with Hoechst. **D** Boxplots represent the median (vertical line) number of mouse sperm binding to mammalian/fish peptide beads or beads alone and data points within the 10th and 90th percentiles (error bars). Boxes include the middle two quartiles, and dots indicate the outliers. Superscript letters show statistical significance (P<0.05) defined by One-Way ANOVA followed by Tukey HSD (honestly significant difference) post-hoc test. **E** Schematic of cross-species insemination assays using mouse sperm to inseminate zebrafish chorion (Top) or eggs (Bottom). **F** Representative pictures of zebrafish chorion or eggs inseminated with mouse sperm; Top-right panel: mouse sperm binding to normal mouse eggs after 60 min of incubation. Inset, 2.0× magnification. Mouse two-cell embryos serve as a negative control for sperm binding. Bottom-right panel: mouse sperm binding to mouse two-cell embryos; mouse eggs serve as an internal positive control for mouse sperm binding. Top-left panel: mouse sperm incubated with zebrafish chorion without ovulated oocyte (60 min incubation). Bottom-left panel: mouse sperm incubated with zebrafish ovulated oocytes (60 min incubation). **G** Boxplots represent the median (vertical line) number of mouse sperm binding to mammalian/fish oocytes or embryos (sperm bound per 20 μm² projected surface area) and data points within the 10th and 90th percentiles (error bars). Boxes include the middle two quartiles, and dots indicate the outliers. Superscript letters show statistical significance (P<0.05) defined by One-Way ANOVA followed by Tukey HSD (honestly significant difference) post-hoc test.

HTF/HSA (mouse IVF conditions) and observed over time (0, 30, 60, 120, and 240 minutes) for morphological signs of deterioration of the chorion. Control eggs ($n = 30$, 3 replicates) were maintained in Hank's solution (Westerfield, 2007) at room temperature (Suppl. Figure 1). Under mouse IVF conditions, zebrafish oocytes presented no signs of zona deterioration within the first 60 minutes (Suppl. Figure 1). Therefore, we used 60 minutes as the time limit for incubating zebrafish chorion or eggs with mouse sperm (previously incubated for 45 min at 37 °C, 5% CO₂ in HTF/HSA).

Zebrafish ovulated eggs were collected and preserved in Hank's saline solution to prevent egg activation (Westerfield, 2007). As HTF contains water, it may induce activation of zebrafish oocytes, which may result in post-fertilization biochemical modification of the ZP proteins of the chorion (Masuda et al., 1991). Thus, we mechanically isolated the chorion from zebrafish eggs in Hank's, moved them to HTF/HSA, and inseminated them with 10⁵ progressive motile mouse sperm (Figure 1E). The isolated chorion was found to not support mouse sperm binding (Figure 1F, top-left). To quantify these observations that involve gametes of considerably different sizes (~80 μm for mouse egg/2-cell embryo vs. ~700 μm for zebrafish egg), we acquired z maximum intensity projections of inseminated eggs/embryos by confocal microscopy (LSM800, Zeiss), and calculated the number of sperm bound per 20-μm² area using the ZEN 3.2 software (Figure 1G). Our quantitative analyses using boxplots showed that mouse sperm bind to the fish chorion (0.025 ± 0.11 , s.e.m.) with significantly lower efficiency to mouse zonae (0.76 ± 0.05 , s.e.m.) (Figure 1F, top-right), and with comparable efficiency to the zonae surrounding mouse 2-cell embryos (0.041 ± 0.01 , s.e.m.) (Figure 1F, bottom-right).

In addition, whole ovulated zebrafish oocytes surrounded by a chorion were placed in HTF/HSA and inseminated with 10⁵ progressive motile mouse sperm (37 °C, 5% CO₂ in HTF/HSA) for 60 minutes. Inseminated zebrafish oocytes showed very few sperm bound to the chorion (Figure 1F, bottom-left). Quantification of these observations found that mouse sperm bind to the chorion surrounding zebrafish eggs with comparable efficiency (0.06 ± 0.02 s.e.m.) as to the chorion only (no oocyte within) isolated from zebrafish eggs in Hank's solution (Figure 1G). From these observations, we conclude that mouse sperm cannot bind to the zebrafish chorion.

To assess whether zebrafish sperm can interact with mammalian eggs, we performed a reciprocal cross-species insemination assay by inseminating MII mouse oocytes with zebrafish sperm under fish insemination conditions. No sperm binding to the zona pellucida was observed (Suppl. Figure 2A and B), suggesting that zebrafish sperm do not recognize or bind to the mammalian egg coat.

Mouse sperm recognize the micropylar region of fish oocytes

Although the chorion does not support mouse sperm binding, the micropyle region of ovulated oocytes is known to attract fish sperm (Yanagimachi, 1957; Yanagimachi et al., 2013). This is due to the presence of still unidentified glycosylated protein(s) present in the micropylar region of fish eggs (Figure 2A and B).

Previous studies have yielded conflicting results regarding the presence of glycoproteins within the zebrafish micropyle. Yanagimachi and collaborators (Yanagimachi et al., 2017) reported the absence of detectable glycoproteins in the micropyle, whereas Dingare et al. (Dingare et al., 2018) identified glycoprotein expression localized to the micropyle region. Given the critical role of the micropyle in directing sperm entry during external fertilization in teleosts, the presence or absence of a specific glycoprotein could have important functional implications in gamete attraction and fertilization. To address this discrepancy, we investigated whether micropyle protein (MP) is indeed present at the zebrafish micropyle and evaluate the potential role of the MP in guiding or facilitating sperm entry.

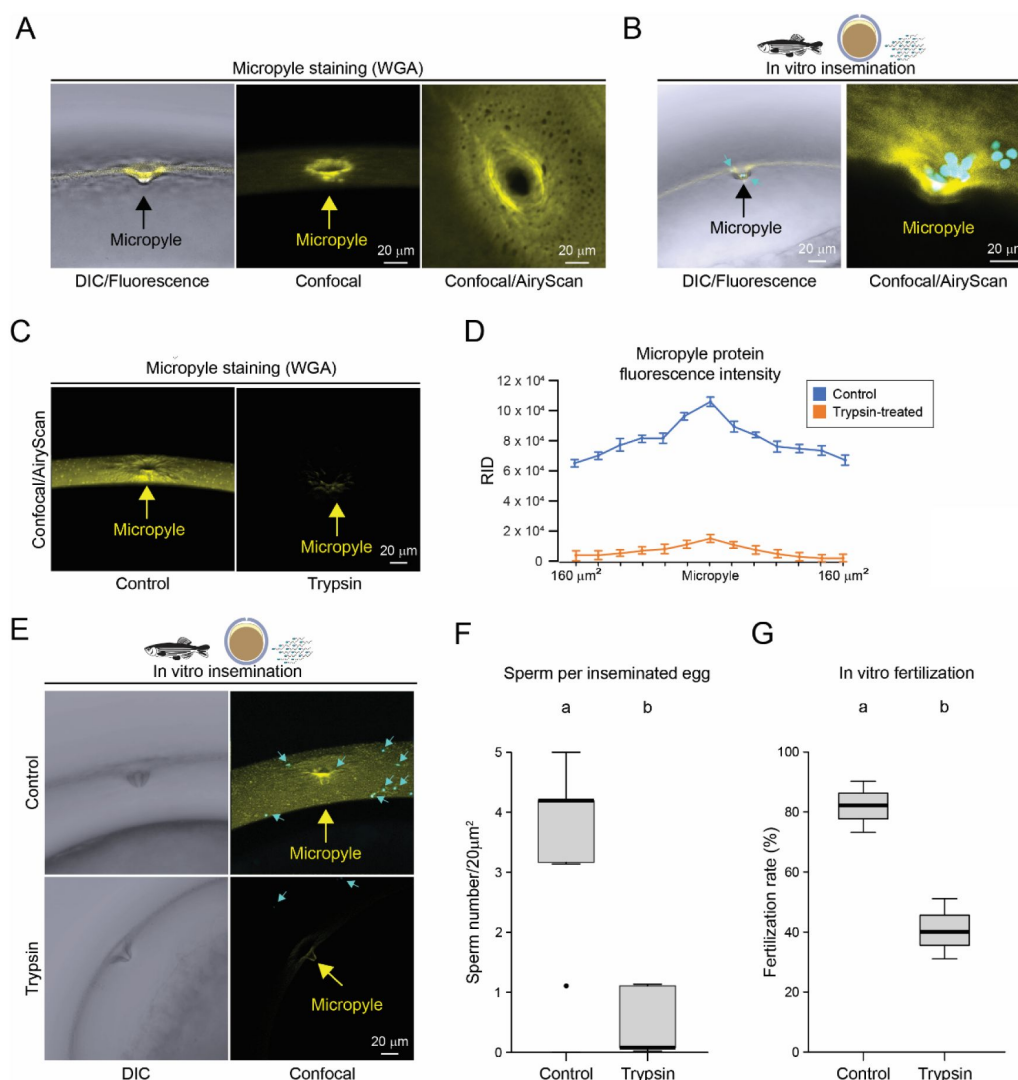


Figure 2.

Fish sperm interaction with the micropyle.

A) Confocal images of the micropyle stained by WGA-633 in zebrafish oocytes ($n > 15$); arrows indicate the micropyle. **B)** Zebrafish *in vitro* insemination: Hoechst-stained zebrafish sperm (light blue) that have approached or entered the micropyle in freshly ovulated oocytes (yellow, WGA-633-stained); samples were fixed in paraformaldehyde few seconds after insemination. **C)** Zebrafish eggs untreated (left) or treated with trypsin to eliminate the micropyle protein (right). **D)** Fluorescence was measured (Fiji/ImageJ) within a $20 \mu\text{m}^2$ area; 0 indicates the micropyle opening position (yellow), $160 \mu\text{m}$ indicates the most distant position measured from micropyle opening. **E)** Same as in (C); left, DIC, right, confocal images (maximum intensity projection) of zebrafish oocytes inseminated with zebrafish sperm. **F)** Same as in 1G, for the quantification of the number of zebrafish sperm approaching and entering the micropyle of oocytes treated/not treated (control) with trypsin ($n = 3$). **G)** Same as in 1G, for fertilization rates ($n = 3$).

To test for the presence of the MP in zebrafish, we stained freshly ovulated oocytes with fluorescently conjugated wheat germ agglutinin (WGA-633). We confirmed the presence of a positive signal in the micropylar region of zebrafish oocytes (**Figure 2A**), as previously described (Dingare et al., 2018). Using fish IVF conditions, we inseminated zebrafish ovulated eggs with zebrafish sperm and imaged sperm around the micropylar area and within the inner opening of the micropyle (**Figure 2B**). The presence of sperm after insemination was found to depend on the presence of the MP. Indeed, when the MP was biochemically removed with trypsin (**Figure 2C** and **D**), the number of zebrafish sperm surrounding the micropylar region or entering the micropyle significantly decreased (**Figure 2E** and **F**), which also led to a reduction in the fertilization rates (**Figure 2G**).

While performing our cross-species insemination assay, upon *in vitro* insemination of zebrafish ovulated oocytes with mouse sperm (37 °C, 5% CO₂, HTF/HSA), we could observe the accumulation of mouse sperm to the chorion area surrounding the micropyle (**Figure 3A**, left panel). This binding persisted after extensive washes with a glass pipette, which was able to detach loosely bound mouse sperm to mouse 2-cell embryos (negative control; **Figure 1F**).

Then, to assess the ability of the zebrafish micropyle to support mouse sperm binding independently from the oocyte, we mechanically isolated the chorion in Hank's solution to prevent oocyte activation. The micropylar region of the isolated chorion supported mouse sperm binding (**Figure 3A**, right panel). The binding pattern of the mouse sperm was comparable with zebrafish sperm binding around the micropyle region (**Figure 2B**). Upon quantification of these observations with boxplots, we found that mouse sperm bound the micropylar region (1.11 ± 0.1 , s.e.m.) with comparable efficiency to mouse zonae pellucidae (1.07 ± 0.07 , s.e.m.) surrounding ovulated eggs (**Figure 3B**), while sperm bound to the chorion away from the micropyle (at a radial distance greater than 160 μm from the opening of the micropyle) were rarely observed (0.2 ± 0.04 , s.e.m.). Trypsin digestion of the chorion virtually eliminated the MP as shown by the absence of WGA-633 staining (**Figure 2C**). When inseminated with mouse sperm, trypsin-treated micropyle regions did not support sperm binding (**Suppl. Figure 3A** and **B**). From these observations, we concluded that mouse sperm can recognize the fish micropyle *in vitro*, and such recognition is dependent on the presence of the MP.

It is conceivable that, if the MP exerts a chemotactic effect on zebrafish and mouse sperm, it would be expected that this factor produces a concentration gradient. To measure if a fluorescence gradient may exist from the periphery towards the micropyle inner opening, via ImageJ/Fiji, we measured the raw integrated density (RID) (Brazill et al., 2018) of the zebrafish WGA-633-stained chorion encompassing the micropyle ($n = 10$) (**Figure 3C**). We found that fluorescence density increases from the periphery towards the central 40 μm area encompassing the micropyle region (**Figure 3D**, left-Y axis). Of note, we observed an increasing number of mouse sperm bound to the chorion from the periphery towards the micropyle (**Figure 3D**, right-Y axis). The direct correlation between the increasing fluorescence density and the number of sperm bound from the periphery towards the micropyle is consistent with the potential role of the MP as a chemoattractant to guide sperm towards the micropyle (Yanagimachi, 1957; Yanagimachi et al., 2013).

To capture live mouse sperm entering the micropyle, we inseminated 20 zebrafish ovulated eggs in 300 μl HTF (Merck Millipore, USA) with 3,000 progressive motile sperm and performed 4 different independent experiments using mouse sperm from three different fertile males. Mouse sperm reached the zebrafish oocytes 48.5 ± 10.6 seconds post insemination and entered the micropyle canal within 24.5 ± 7.1 seconds after approaching the zebrafish oocyte, attempting to cross the micropylar opening (**Figure 3E**, **Suppl. Figure 4** and **Suppl. Videos 1-2**). Under mouse IVF conditions, the micropyle detached 6–7 minutes after imaging started and continued progressively over time (**Suppl. Figure 5**).

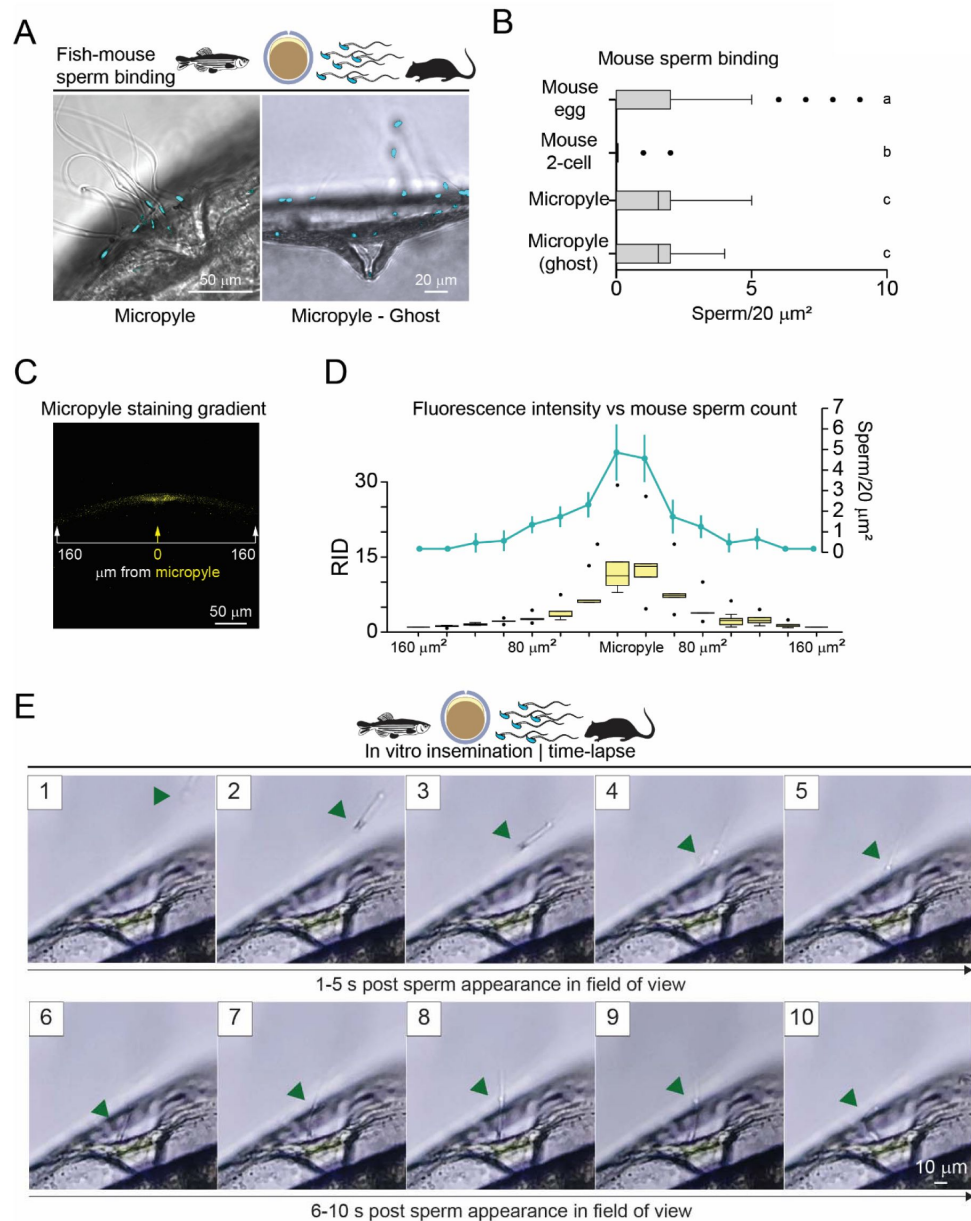


Figure 3.

Mouse sperm cross the fish micropyle.

A Cross-species insemination: mouse sperm (Hoechst-stained, light blue) in the zebrafish micropyle region of a chorion (60 min incubation) surrounding the oocyte (left), or of a chorion mechanically freed from the oocyte (right, Ghost). **B** Quantification of mouse sperm in the micropyle region of chorion with or without zebrafish oocyte: same as 1G; mouse eggs or two-cell embryos served as an internal positive and negative control for sperm binding ($n = 3$). **C** X-Y plane confocal projection of zebrafish chorion encompassing the WGA-633 (yellow) micropyle region ($\sim 320 \mu\text{m}$); bar with arrows indicates positions at which fluorescence was measured (Fiji/ImageJ) as in 2D. **D** Quantification of mouse sperm across the micropyle region: boxplots represent the median (vertical line) raw integrated density ratio (RID ratio, left Y axis) measured on ten zebrafish chorions; data points within the 10th and 90th percentiles (error bars). Boxes include the middle two quartiles, and dots indicate the outliers. A light blue line represents the number of sperm (right Y axis) found in the corresponding chorion position (X axis); error bars represent s.e.m.; statistical significance ($P < 0.05$) across RID ratios or sperm numbers in different positions is defined by One-Way ANOVA followed by Tukey HSD post-hoc test. **E** Time-lapse frames from supplementary video 1, showing the first sperm entering the micropyle of a freshly ovulated zebrafish egg. Green arrowheads indicate mouse sperm. Insets show the number of seconds ('s') after the first sperm appears in the field of view.

Mouse sperm cross the fish micropyle opening

In mammalian fertilization, once sperm bind to the zona pellucida, they cross it to reach the perivitelline space (PVS), a necessary step prior to fusion with the oolemma. Previous studies using cross-species insemination assays have shown that when mammalian sperm are able to bind the egg coat of another species, they can cross its extracellular matrix (Baibakov et al., 2012 [↗](#); Yauger et al., 2011 [↗](#)). For instance, human sperm have been shown to cross the zona pellucida of transgenic mouse eggs expressing human ZP2 and accumulate within the PVS, as human sperm do not fuse with the mouse oolemma (Baibakov et al., 2012 [↗](#)). Given our observation that mouse sperm interact with the micropyle region and are capable of entering the micropyle canal, we investigated whether they could similarly cross the micropyle and reach the ICS.

Using confocal microscopy (Zeiss LSM 800), we found that multiple mouse sperm (115.87 ± 0.01) entered the micropyle canal and crossed the micropyle opening, accumulating in the ICS (**Figure 4A** [↗](#), **Suppl. Figure 6** [↗](#); $n = 12-14$ eggs, 3 replicates), representing $0.11587 \pm 0.0001\%$ of the total inseminated progressively motile sperm. After crossing, sperm concentrated either within $160 \mu\text{m}$ radius around the micropyle opening (**Figure 4A** [↗](#) left and mid panels) or away from the micropyle (farther than $160 \mu\text{m}$ from the micropyle opening; **Figure 4A** [↗](#), right panel). Quantifications of these observations revealed a comparable, yet significantly higher, number of ICS mouse sperm around the micropyle than away from the micropyle (**Figure 4B** [↗](#)).

We, then, wondered at what site of the micropyle region mouse sperm crossed the chorion. Previous studies have shown that, after zona binding, mammalian sperm can cross the matrix at any binding site (VandeVoort et al., 1997 [↗](#); Wakayama et al., 1996 [↗](#); Yanagimachi & Phillips, 1984 [↗](#)), whereas fish sperm cross the chorion only at the micropyle inner opening (Yi et al., 2019 [↗](#)). Thus, we investigated whether mouse sperm cross the chorion at any site of the micropyle region or, like fish sperm, use the inner opening of the micropyle. Using scanning electron microscopy (SEM), we observed mouse sperm within the micropyle canal (**Figure 4C** [↗](#), left). However, no mouse sperm were found mid-way through the chorion in the micropyle region, indicating that mouse sperm, likely, used the micropyle inner opening to cross the fish chorion. To validate this thesis, we used transmission electron microscopy (TEM) to study mouse sperm in the micropyle canal (**Figure 4C** [↗](#), mid and right). TEM analyses confirmed that mouse sperm did not cross the chorion at the binding site. From this, we concluded that the interaction between mouse sperm and the zebrafish micropyle may differ from the mechanism of sperm binding with the zona pellucida. To that end, we used sperm from transgenic mice to further characterize the nature of this interaction.

Ineffective induction of mouse sperm acrosome exocytosis during interaction with the fish micropyle

In mammals, acrosome exocytosis is a necessary step in fertilization. A fusion event between the plasma membrane and the outer acrosomal membrane enables the release of acrosomal contents and exposes the inner acrosomal membrane along with the equatorial segment—a remnant of the sperm plasma membrane—which is necessary for subsequent fusion with the oolemma. In contrast, sperm from most teleost species, including zebrafish, lack an acrosome entirely, and thus do not rely on this mechanism for fertilization. However, in a few chondrosteian fish species such as sturgeon, acrosome-bearing sperm undergo acrosome exocytosis specifically within the micropyle canal just prior to gamete fusion (Psenicka et al., 2010 [↗](#)). The absence of an acrosome in zebrafish sperm suggests that their micropyle may not possess the molecular signals required to induce this process (Hirohashi & Yanagimachi, 2018 [↗](#)). While such a conclusion is plausible, it has not been directly tested. To address this, we examined whether the zebrafish micropyle can elicit acrosome exocytosis in mouse sperm, which indeed, present an acrosome. This experiment was designed to determine whether the micropyle microenvironment provides inductive cues sufficient to trigger this exocytotic event or whether acrosome exocytosis is strictly mediated by

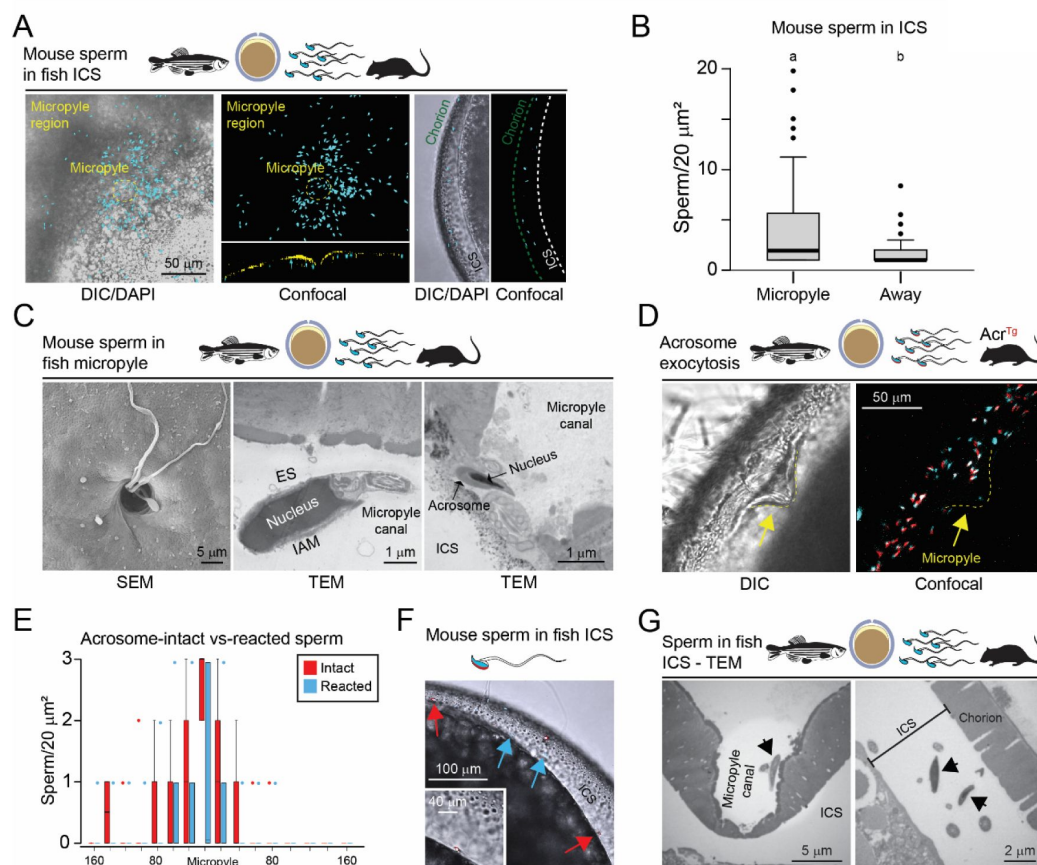


Figure 4.

Localization and dynamics of sperm interactions with the micropyle and inter-chorion space in zebrafish oocytes.

A) Hoechst-stained sperm (blue) which has crossed the micropyle. DIC (left) and confocal (mid-panel) projection of sperm accumulated in the inter-chorion space (ICS) of a zebrafish oocyte imaged from the top; in the left- and middle/top panels, the 633 channel is turned off to visualize the sperm accumulated in the ICS around the micropyle region (yellow circled); inset shows a longitudinal section of the same oocyte, showing the Hoechst-stained sperm (blue) under the WGA-633-stained micropyle region (yellow). Right, Hoechst-stained sperm (blue) is included in the ICS. **B**) Quantification as in 1D of mouse sperm in the ICS below (left) or away (right) from the micropyle region. Error bars represent s.e.m., statistical significance ($P < 0.05$) is defined by One-Way ANOVA followed by Tukey HSD. **C**) Electron microscopy of mouse sperm in the zebrafish micropyle region and within the micropyle canal, 1 hour after insemination. Left, SEM, mid and right panels, TEM; IAM, inner acrosomal membrane. OAM, outer acrosomal membrane; ES, equatorial segment. **D**) Acr^{Tg} sperm in the micropyle region. Left, DIC, right, confocal projection. The yellow arrow indicates the micropyle opening. **E**) As in 1G, with Acr^{Tg} sperm (intact vs. reacted). **F**) Acrosome-intact (red arrows) and reacted (light-blue arrows) mouse sperm in zebrafish ICS. Inset represents a 2x magnification of an acrosome-intact sperm in the ICS. **G**) Left panel, micropyle structure after the entry of multiple mouse sperm. Right panel, acrosome-intact sperm from the ICS of the oocyte, the micropyle of which is shown in the left panel. Black arrows, mouse sperm.

species-specific gamete interactions. To determine whether the zebrafish micropyle can induce mouse sperm acrosome exocytosis, we used TEM and confocal microscopy to track the status of the acrosomes of mouse spermatozoa during cross-species insemination. We used transgenic mouse sperm that accumulate mCherry in the acrosome (Avella et al., 2016 [DOI](#)) (*Acrosin*^{Tg}). Acrosome-intact sperm present a visible fluorescent acrosome beneath the nucleus (Avella et al., 2016 [DOI](#)). During acrosome exocytosis, mCherry is released upon fusion of the sperm plasma membrane with the outer acrosomal membrane (7). This results in the loss of fluorescent signal and marks these sperm as ‘acrosome-reacted’, as documented by fluorescent and electron microscopy analyses (Avella et al., 2016 [DOI](#)). Through confocal microscopy, we found $86 \pm 4\%$ ($n = 10$, s.e.m) of sperm surrounding the micropyle to be acrosome intact (**Figure 4D** [DOI](#) and **E** [DOI](#)). Sperm could cross the micropyle inner opening irrespective of their acrosomal status (reacted or intact; **Figure 4C** [DOI](#), mid and right, respectively). Additionally, both acrosome-intact and acrosome-reacted sperm were found in the ICS via confocal and TEM (**Figure 4F** [DOI](#) and **G** [DOI](#)), though they were never observed to fuse with the egg cell under our experimental conditions (**Figure 4D** [DOI](#)), possible due to predicted low affinity between the Izumo1-Spaca6-Tmem81 mouse sperm trimeric complex (Deneke et al., 2024 [DOI](#)) with the zebrafish egg cell protein Bouncer (**Suppl. Figure 7** [DOI](#)). Thus, we concluded that mouse sperm can recognize the micropylar region of zebrafish oocytes and cross the micropyle inner opening regardless of the acrosome status, though passage through the micropyle does not appear to induce acrosome exocytosis.

Hyperactive motility is necessary to mediate sperm crossing the zebrafish micropyle

We found that mouse sperm in the fish ICS were still motile and presenting what may appear to be a hyperactive motility pattern (Suppl. Videos 3-5).

In mammals, hyperactivation is necessary for crossing the zona (Miller, 2024 [DOI](#)). To investigate whether hyperactivated motility is also required for mouse sperm to locate or traverse the zebrafish micropyle, we performed cross-species insemination experiments using sperm from *CatSper1*^{Null} mice. *CatSper1* encodes a pore-forming subunit of the sperm-specific CatSper Ca²⁺ channel, which is necessary for the acquisition of an asymmetric, high-amplitude flagellar bending characteristic of hyperactivation. *CatSper1*^{Null} sperm are unable of crossing the zona, with prior studies typically inferring this defect from the absence of fertilization (Ren et al., 2001 [DOI](#)).

CatSper1^{Null} sperm are known to progressively lose motility over time (Qi et al., 2007 [DOI](#)) and are unable to undergo hyperactivation, resulting in failed zona penetration and infertility (Ren et al., 2001 [DOI](#)). Because impaired motility could potentially affect sperm binding to the zona, we first sought to quantitatively evaluate whether *CatSper1*^{Null} sperm were able to bind to the zona upon incubation under capacitating conditions. Thus, first, we quantified the ability of *CatSper1*^{Null} sperm to bind to mouse zonae *in vitro* or to cross the zonae *in vivo*. To quantify the ability of *CatSper1*^{Null} sperm to bind to the zona matrix, we incubated sperm from infertile *CatSper1*^{Null} males (no sired pups after mating for 1.5 months with fertile females) or fertile *CatSper1*^{Het} males (≥ 1 litter sired after mating with fertile females for 1.5 months) for 45 min in HTF/HSA, 37 °C, 5% CO₂, and used either mutant sperm to separately inseminate fertile ovulated MII eggs that were previously denuded from cumulus masses by hyaluronidase (16-22 eggs, 3 replicates). We found that *CatSper1*^{Null} sperm bound to mouse zonae with comparable efficiency (60.2 ± 3.55 ; s.e.m.) to *CatSper1*^{Het} sperm (66.5 ± 1.26 , s.e.m) (**Figure 5A** [DOI](#) and **B** [DOI](#)).

While it is well known that *CatSper1*^{Null} sperm fail to fertilize eggs, prior studies have not quantitatively assessed their ability to reach the perivitelline space. To address this, we used *Cd9*^{Null} oocytes, which enable direct measurement of sperm accumulation in the PVS and help identify defects in zona penetration. To quantify the ability of *CatSper1*^{Null} sperm to cross the zona, we mated five *CatSper1*^{Null} or *CatSper1*^{Het} males with five *Cd9*^{Null} females, whose eggs rarely fuse

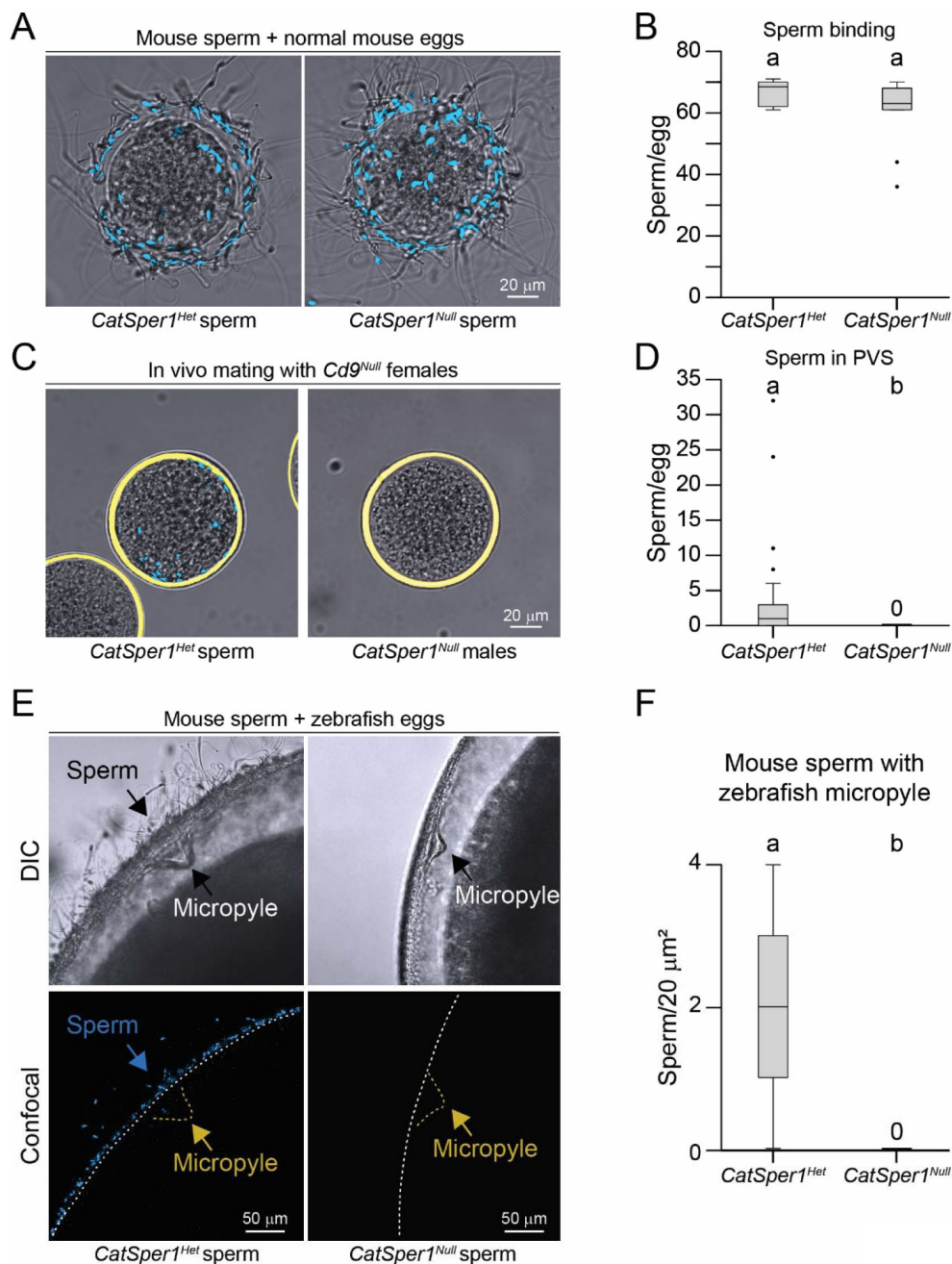


Figure 5.

CatSper is necessary for mouse sperm crossing the zebrafish micropyle.

A) Hoechst-stained fertile *Catsper1^{Het}* (Control, left) and *Catsper1^{Null}* sperm binding to ovulated mouse eggs. **B)** Same as in 1D, per mouse egg. **C)** Fertile *Catsper1^{Het}* (Control, left) and *Catsper1^{Null}* sperm in the PVS of *Cd9^{Null}* female eggs upon *in vivo* mating. **D)** Same as in 1D, number of PVS sperm per *Cd9^{Null}* mouse egg. **E)** Fertile *Catsper1^{Het}* (Control, left) and *Catsper1^{Null}* (right) Hoechst-stained sperm in the micropyle region of zebrafish eggs; top, DIC image, bottom, confocal. **F)** Same as in 1G, with *Catsper1^{Het}* vs. *Catsper1^{Null}* sperm.

with sperm, leading to an accumulation of supernumerary sperm in the PVS. We found that *Cd9^{Null}* eggs from females mated with *CatSper1^{Het}* males accumulated 4.1 ± 1.57 (s.e.m.) in the PVS, whereas there were no sperm detected in the *Cd9^{Null}* eggs from females mated with *CatSper1^{Null}* males (10 females, 10-15 eggs/female) (**Figure 5C** and **D**). These data confirmed the inability of *CatSper1^{Null}* sperm to cross the zona *in vivo* and are consistent with previous reports showing *CatSper1^{Null}* sperm being unable to fertilize eggs under *in vivo* conditions (Ren et al., 2001).

To see whether hyperactive motility is necessary for the mouse sperm to enter and cross the zebrafish micropyle, we inseminated zebrafish eggs with *CatSper1^{Null}*, using *CatSper1^{Het}* sperm as controls. Although *CatSper1^{Null}* sperm are unable of acquiring hyperactivated motility and exhibit a decline in progressive motility over time, previous studies have shown that 20% of *CatSper1^{Null}* sperm retain progressive motility at 90 minutes (Qi et al., 2007). Considering that we inseminated with 100,000 progressively motile sperm, and that 15–35% of capacitated sperm are typically hyperactivated by 90 minutes (Goodson et al., 2011), we inferred that we could have relied on approximately 3,000 *CatSper1^{Null}* sperm with progressive motility, which was indeed sufficient to locate and bind mouse zonae *in vitro* (**Figure 5A** and **B**). Therefore, we performed cross-species insemination, using *CatSper1^{Null}*. We rarely found *CatSper1^{Null}* sperm in the micropyle of zebrafish eggs and never found *CatSper1^{Null}* sperm in the fish ICS, indicating that CatSper is necessary for mouse sperm to enter cross the micropyle (**Figure 5E** and **F**).

Discussion

By establishing a mouse-fish gamete interaction assay, we report that, although the chorion cannot support mouse sperm binding, a subpopulation of mouse sperm can find and enter the fish micropyle. Mouse sperm cross the micropyle and accumulate in the fish PVS. Using sperm from different transgenic mouse lines, we found that the recognition of the micropyle by mouse sperm is dependent on the CatSper channel and that the zebrafish micropyle failed to effectively induce acrosome exocytosis.

By Clustal Ω sequence alignment (Sievers et al., 2011), it appears that zebrafish zona proteins lack an N-terminal domain with conserved sequence identity to the mammalian ZP2 N-terminus (Avella et al., 2014). Therefore, the observed absence of mouse sperm binding to zebrafish ZP2 or the chorion is consistent with a model that envisions the N-terminal region of mammalian ZP2 to be required for mouse sperm binding to an extracellular matrix (Avella et al., 2014; Bhakta et al., 2019). Of note, consistent with the hypothesis that zebrafish sperm do not recognize or interact with mammalian zonae, we did not observe any zebrafish sperm bound to the mouse zona pellucida (**Suppl. Figure 2**). These results highlight that cross-species insemination assays do not necessarily exhibit reciprocal sperm-egg interactions. For instance, human sperm bind specifically to human zonae but not to those of mice, whereas mouse sperm are capable of binding to both mouse and human zonae (Avella et al., 2014; Baibakov et al., 2012; Bedford, 1977). Such asymmetry may reflect evolutionary divergence and species-specific specialization in gamete recognition mechanisms.

A model of gamete recognition at the zona pellucida suggests that mouse sperm can cross the extracellular matrices provided they are able to bind, and this binding is regulated by the cleavage status of ZP2. In unfertilized eggs, ZP2 is uncleaved, supporting sperm binding and subsequent penetration. Following gamete fusion, cortical granule exocytosis releases the metalloprotease ovastacin, which cleaves the N-terminus of ZP2, rendering the zona unable to support additional sperm binding and thereby preventing further penetration (Bhakta et al., 2019). Notably, fertilized eggs or two-cell embryos that retain uncleaved ZP2 continue to support *de novo* sperm binding, allowing supernumerary sperm to traverse the zona and accumulate in the perivitelline space, unable to fuse with the oolemma or blastomeres (Baibakov et al., 2007, 2012; Burkart et al., 2012; Gahlay et al., 2010). Analogously, in our experimental system, mouse sperm were

observed to interact with the zebrafish micropyle opening, and this interaction was as stable as the sperm binding to the zona after washes with 2-cell embryos controls. We interpret these findings to suggest that, once the interaction at the micropyle occurs, mouse sperm can cross the chorion, even under conditions in which the micropyle may have begun detaching from the oocyte due to activation. These observations imply that mouse sperm may circumvent the zebrafish oocyte's mechanism for restricting supernumerary sperm crossing the micropyle, even after egg activation has occurred.

The ability of sperm to effectively localize eggs is vital for external fertilizing species that emit their gametes in the water column. In sea urchins, small peptides such as Resact have been identified as the molecule mediating the sperm attracting stimulus to the unfertilized egg. In zebrafish, and other teleost fish species, sperm contact with water is sufficient to activate their motility, yet as they pass near the micropyle area, their motility increases (Suzuki, 1958). Such a gain in motility appears to be elicited by exposure to the MP that guides the fertilizing sperm into the micropyle (Yanagimachi et al., 1992). We and others have shown that removal of the MP by trypsin digestion significantly decreases egg fertilization rate, indicating the important role of MP in fish fertilization (Yanagimachi et al., 2013). Sea urchin Resact induces a cGMP signaling pathway that mediates Ca^{2+} bursts via the CatSper channel, and the increased cytoplasmic Ca^{2+} concentrations confer sperm chemotactic steering towards the egg (Seifert et al., 2015). In addition, it has been hypothesized that the fish CatSper channel may also play a role in activating fish sperm motility during fertilization (Yanagimachi et al., 2017). Upon binding with motile sperm, the MP may activate sperm proton pumps, leading to an increase in intracellular pH; this, in turn, would activate CatSper channels, resulting in a Ca^{2+} influx and higher levels of intracellular Ca^{2+} . Higher concentrations of intracellular Ca^{2+} would make the sperm tail beating pattern change from symmetrical to asymmetrical, directing sperm entry into the micropyle (Yanagimachi et al., 2017). Consistent with these hypotheses, our data demonstrates that CatSper is necessary to mediate mouse sperm entering and crossing the zebrafish micropyle. Like fish sperm, it is conceivable that the MP may elicit an attractant stimulus towards a subpopulation of mouse sperm. However, recent findings have shown successful cross-species fertilization of evolutionary-distant fish species, such as with female common carp and male blunt snout bream (S. Wang et al., 2017) or Russian Sturgeon with American Paddlefish (Káldy et al., 2020), suggesting that such a stimulus may not act with species-specificity. This is consistent with findings that show how sperm from other fish species (Yanagimachi et al., 2013, 2017) may reach and cross the micropyle of different fish species (Yanagimachi et al., 2017).

While our results show an absence of *CatSper1*^{Null} sperm accumulation within the micropyle and ICS, we acknowledge that this phenotype could be interpreted also as a consequence of general motility defects rather than a specific failure in hyperactivation or micropyle recognition. *CatSper1*^{Null} sperm are indeed characterized by an inability to develop hyperactivated motility and by a gradual decline in progressive motility over time, with only ~20% of sperm remaining motile after 90 minutes compared to over 70% in controls (Qi et al., 2007). However, under our IVF conditions, *CatSper1*^{Null} sperm maintained sufficient progressive motility within the first hour post-insemination to bind the zona with efficiency comparable to *CatSper1*^{Het} controls. Given previous reports that 15–35% of sperm achieve hyperactivation by 90 minutes (Goodson et al., 2011), and considering our use of 100,000 progressively motile sperm per insemination, we estimate that approximately 3,000 hyperactivated *CatSper1*^{Null} sperm were present in the dish. Despite this, no sperm were observed in proximity to the micropyle or within the ICS. These findings support the interpretation that failure to hyperactivate, rather than total loss of motility, is the key factor preventing *CatSper1*^{Null} sperm from locating and traversing the micropyle. In addition, the inability of *CatSper1*^{Null} sperm to locate or traverse the micropyle may stem from defective rheotactic responses, which are critical for guiding sperm toward chemoattractant gradients (Chung et al., 2017).

The potential existence of a mammalian sperm chemoattraction mechanism has been the focus of considerable research efforts. Using *in vitro* assays to quantify chemoattraction in mammalian sperm (e.g., Zigmond chamber), several studies have shown evidence of a putative attraction mechanism coming from the egg microenvironment (Eisenbach & Giojalas, 2006 [↗](#)), with subpopulations of sperm that responded to chemical gradients obtained from egg extracts or oviductal fluids (Fabro et al., 2002 [↗](#); Giojalas & Rovasio, 1998 [↗](#); Oliveira et al., 1999 [↗](#); Sun et al., 2005 [↗](#)). However, no follow-up studies with genetic loss-of-function assays could characterize such a mechanism further. Hence, it is still uncertain whether mammalian sperm may respond to attracting stimuli from an egg (Yanagimachi, 2022 [↗](#)). Our data suggests the presence of a conserved mechanism that attracts mouse sperm towards eggs. Although such stimulus comes from a fish egg, it is plausible to reason that a mammalian protein homologous to the fish MP may play a similar role towards a subpopulation of mouse sperm that swim near an ovulated egg in the ampulla. Unfortunately, studies using mass spectrometry, together with cloning, immunohistochemistry, and *in situ* hybridization, were unable to identify the nature of the fish MP (Oda et al., 1995 [↗](#), 1998 [↗](#); Vines et al., 2002 [↗](#), p. 202). Therefore, this factor is still unidentified.

To successfully fertilize an egg, mammalian sperm must undergo acrosome exocytosis, as only acrosome-reacted sperm can fuse with the oolemma (Bhakta et al., 2019 [↗](#)). The acrosome is a Golgi-derived organelle that underlies the anterior plasma membrane heads; it is enclosed by an inner and outer membranes and containing digestive enzymes (none found necessary for zona pellucida) (Bhakta et al., 2019 [↗](#); Buffone et al., 2014 [↗](#)). During capacitation, sperm undergo the acrosome reaction, a membrane fusion event that exposes the inner acrosomal membrane and equatorial segment. Fusion with the oolemma occurs via remnants of the sperm plasma membrane overlying the equatorial segment (Yanagimachi & Noda, 1970 [↗](#)). Typically, fish sperm do not present an acrosome. However, in certain fish species such as sturgeon, sperm possess an acrosome and undergo acrosome exocytosis while traversing the micropyle before gamete fusion (Alavi et al., 2012 [↗](#); Psenicka et al., 2010 [↗](#)). In contrast, zebrafish sperm lack an acrosome, suggesting that the zebrafish micropyle may not have evolved mechanisms to trigger acrosome exocytosis. Consistent with this, we observed acrosome-intact mouse sperm both within the micropyle and in the ICS, indicating that passage through the zebrafish micropyle does not induce acrosome exocytosis.

In our studies, mouse sperm found in the micropyle or the PVS ICS of zebrafish eggs were often observed to be acrosome-intact. Failed acrosome exocytosis has been observed in previous cross-species insemination studies in rodents, where sperm could penetrate zonae of different species while remaining acrosome-intact (VandeVoort et al., 1997 [↗](#); Wakayama et al., 1996 [↗](#); Yanagimachi & Phillips, 1984 [↗](#)), indicating putative species-specific oocyte mechanisms of induction of acrosome exocytosis (Avella & Dean, 2011 [↗](#)). As zebrafish sperm do not present an acrosome, it is conceivable that, unlike sturgeon, zebrafish oocytes may lack the ability to induce acrosome exocytosis in the micropyle.

While this study does not aim to model physiological fertilization, our findings offer insight into potential evolutionary conservation in gamete guidance mechanisms. Despite substantial evolutionary divergence between zebrafish and mice, the possibility of conserved molecular pathways guiding gamete interaction remains plausible. For example, the CatSper calcium channel, which plays a central role in mediating sperm hyperactivation, is evolutionarily conserved and functionally critical across diverse taxa, from sea urchins (external fertilizers) (Seifert et al., 2015 [↗](#)) to mammals (internal fertilizers) (Lishko & Mannowetz, 2018 [↗](#)). Similarly, although zebrafish sperm lack an acrosome, other fish species such as sturgeon have an acrosome that undergoes exocytosis as sperm traverse the micropyle (Psenicka et al., 2010 [↗](#)). In ovoviparous fish such as black rockfish, sperm undergo complex physiological adaptations within the female reproductive tract—including glycocalyx remodeling and immunomodulatory changes—that contribute to fertilization efficiency (Li et al., 2024 [↗](#)). In mammals, capacitation

encompasses the acquisition of hyperactivated motility and the competence to undergo acrosome exocytosis, both of which are essential for fertilization (Bhakta et al., 2019 [DOI](#); Puga Molina et al., 2018 [DOI](#); Yanagimachi, 1957 [DOI](#); Yanagimachi et al., 2017 [DOI](#)). Notably, features functionally similar to capacitation have also been reported in external fertilizers. In the Pacific herring, sperm remain quiescent upon release into seawater and initiate motility only upon nearing the micropyle, likely in response to egg-derived signals (Yanagimachi, 1957 [DOI](#); Yanagimachi et al., 2017 [DOI](#)). In other teleost such as bitterling and zebrafish, sperm exhibit enhanced motility near the micropyle region (Suzuki, 1958 [DOI](#); Yanagimachi et al., 2017 [DOI](#)). These findings collectively support the notion that sperm-egg guidance cues and changes in motility are conserved across species with distinct fertilization ecologies.

The observation that mouse sperm can locate and traverse the zebrafish micropyle, despite the absence of species-specific zona binding and the stark divergence in fertilization strategies, suggests the presence of conserved cues recognized by mammalian sperm. These results support the thesis that mechanisms regulating gamete guidance may be shared across vertebrates. The molecular identity of such attractants remains unknown, and future studies aimed at identifying these factors will be essential to better understand the mechanisms underpinning gamete attraction across species.

In addition, the existence of a MP mediating fish and mouse sperm raises the prediction for the potential existence of a mouse ortholog for the fish MP and of a cognate sperm receptor. Future studies will focus on identifying the fish MP, looking for mammalian orthologs, and characterizing its expression during fish and mouse oogenesis. In addition, mutant-null fish or mouse models will aid in characterizing the role(s) of these proteins in regulating fish and mouse fertilization.

Materials and Methods

IACUC sentence

Experiments with zebrafish (AB line) or normal and transgenic mice were performed in compliance with the guidelines of the Animal Care and Use Committee of Sidra Medicine and the University of Tulsa under the approved animal study protocols, Sidra Medicine IACUC 2204916, and University of Tulsa TU-0050 and TU-0050R1

Genotyping

Transgenic mice were genotyped by PCR ([95°C for 30 s, 56°C for 30 s, 72°C for 1 min] × 30 cycles, 72°C for 7 min, and 4°C for >30 min) using primers from previous publications *Acr*^{Tg} (Avella et al., 2016 [DOI](#)), *CatSper1*^{Null} (Ren et al., 2001 [DOI](#)), and *Cd9*^{Null} (Le Naour et al., 2000 [DOI](#)).

Light microscopy

Samples were mounted in PBS, and images of oocytes, embryos, and beads were acquired with a dissecting microscope SMZ-2B (Nikon, Japan) at 5X magnification, with an inverted microscope ECLIPSE TS100 (Nikon) or with a confocal microscope (LSM 800, Zeiss) using 10X or 20X objective lens at room temperature (Avella et al., 2014 [DOI](#)). Using ZEN 3.2 image software (Zeiss), LSM 800 images were exported as 300 dpi resolution TIF files and combined using Adobe Illustrator 25.0. Alternatively, maximum intensity projections of confocal optical sections to a single plane were acquired and merged with DIC images of oocytes, embryos, or agarose beads.

Electron microscopy

Zebrafish oocytes inseminated with mouse sperm were fixed in 2% glutaraldehyde in 0.1 M cacodylate buffer (pH 7.4) and incubated at 4°C for 2 hours. After a series of washes in cacodylate buffer, the eggs were embedded in 2% agarose. The samples were dehydrated through a graded series of ethanol and processed for embedding in LR White resin. Ultrathin sections were obtained with an ultramicrotome (Microm International GmbH) and mounted on formvar-coated nickel grids (Electron Microscopy Sciences). Ultrathin sections were counterstained with uranyl acetate followed by lead citrate and imaged in a Jeol JEM-1011 transmission electron microscope (Jeol).

Expression of recombinant zebrafish ZP2 and ZP3

cDNA encoding zebrafish ZP2 (AAK16577.1; 25-405 aa) and ZP3 (NP_571406; 22-396 aa) (H. Wang & Gong, 1999 [\[1\]](#)) were cloned into pFastBac-HBM TOPO (Invitrogen) downstream of a polyhedron promoter and a gp67 signal peptide from *Autographa californica* nuclear polyhedrosis virus (AcMNPV) (38 aa). At the 3' end of the open reading frame, each clone carried a 6-His tag encoded in frame to enable peptide purification. Bacmid DNA, isolated after transformation into DH10Bac *Escherichia coli*, was used to transfect insect cells to produce recombinant baculovirus particles for infection of Sf9 cells using manufacturer protocols (Genscript). Recombinant peptides were purified on IMAC Sepharose High-Performance beads (Cytiva) per manufacturer instructions and assayed on SDS-PAGE as previously reported (Avella et al., 2014 [\[2\]](#)). Plasmid cloning and peptide expression were performed by Genscript (USA).

Mouse gamete collection and *in vitro* insemination

Female mice were stimulated with 5 IU of equine chorionic gonadotropin (eCG) (ProspecBio) and 48 hrs later with human chorionic gonadotropin (hCG) (Sigma-Aldrich). Twelve hours after hCG injection, eggs in cumulus were collected and incubated (37 °C, 5% CO₂) in HTF/HSA under mineral oil before insemination with mouse sperm. Alternatively, eggs were denuded from the cumulus mass with hyaluronidase (Millipore) and washed in HTF/HSA before insemination with mouse sperm. Fertilization was scored 24 hours post-insemination by quantifying the number of two-cell embryos. To assess *in vivo* female fertility, mutant female mice (n ≥ 3) were co-caged with one control fertile female and one fertile male mouse, and litters were recorded until females gave birth to at least three litters.

Epididymal sperm from 10-14 weeks old male mice (normal or *Catsper1*^{Het} and *Catsper1*^{Null}) were incubated under capacitating conditions in human tubal fluid (HTF) supplemented with 0.4% Human Serum Albumin (HSA; Cooper Surgical) for 45 min (37 °C, 5% CO₂) (Avella et al., 2014 [\[2\]](#)). Sperm were added to oocytes or embryos in 50 µl of HTF/HSA drops under mineral oil at a final concentration of 10⁵/ml progressive motile sperm quantified by hemocytometer. Upon coincubation, inseminated oocytes/embryos were washed by gently pipetting with a 200 µl microcapillary pipette (Cooper Surgical) to remove loosely bound sperm. Samples were, then, fixed in 2% paraformaldehyde and stained with Hoechst and WGA-633 (ThermoFisher) to identify the nuclei and the glycoproteins of the zona pellucidae. To assess mouse sperm binding, mouse eggs and two-cell embryos were used as positive and negative controls, respectively.

Zebrafish gamete collection and *in vitro* insemination

Zebrafish females or males (separated from tank mates the night prior) were anesthetized with tricaine (3-amino benzoic acidethylester) solution (Sigma). After being gently dried with a kimwipe, each female was moved to a dry petri dish, and mild pressure on the fish belly was applied, stroking from anterior to posterior. Released oocytes were separated from the female using flat forceps and inseminated with zebrafish sperm. Alternatively, zebrafish oocytes were preserved in Hank's saline (Westerfield, 2007 [\[3\]](#)) and, then, were gently acclimated to HTF/HSA (Cooper Surgical) and inseminated with mouse sperm previously incubated under capacitating

conditions (HTF/HSA, 37 °C, 5% CO₂) (Avella et al., 2014 [↗](#)). Before insemination with mouse sperm, zebrafish oocytes were, in some instances, treated with 0.001% trypsin (crystalline, VWR) in Ringer's solution (Westerfield, 2007 [↗](#)) for 1 – 5 min, and rinsed twice in Hank's solution before cross-species insemination (HTF/HSA, 37 °C, 5% CO₂). Zebrafish oocytes and mouse sperm were incubated for 60 min (HTF/HSA, 37 °C, 5% CO₂), washed in HTF/HSA to remove loosely bound sperm, and video-imaged or fixed in 2% paraformaldehyde. Zebrafish males were dried with a kimwipe, and both sides of the body were firmly stroked to collect sperm out of their genital pore with a microcapillary pipette (Cooper Surgical). Sperm were preserved in ice-cold, full-strength Hank's saline until zebrafish or mouse oocyte insemination (Westerfield, 2007 [↗](#)). Upon oocyte insemination, samples were immediately fixed in 2% paraformaldehyde and stained with Hoechst and Wheat Germ Agglutinin, Alexa Fluor™ 633 Conjugate (WGA-633, ThermoFisher) to identify the nuclei and the micropyle, respectively. Fluorescence intensity of the micropyle was measured as Raw integrated density using Fiji/ImageJ (Brazill et al., 2018 [↗](#)).

Live imaging of mouse sperm attraction to the micropyle of zebrafish mature eggs

Ovulated zebrafish oocytes were collected, acclimated as described above, and immediately transferred to the imaging dish containing 300 µl of HTF/BSA under mineral oil. The micropyle area was identified and brought into focus. Mouse sperm, collected and capacitated as previously described, were prepared for imaging. Images were captured using an inverted microscope (Nikon Eclipse Ti) equipped with a 12MP camera. After focusing on the micropyle, 3,000 progressively motile sperm were introduced into the imaging dish, and time-lapse live imaging was performed to record gamete interactions.

Gamete or embryo quantifications and statistics

Mouse sperm bound to oocytes or beads were quantified from z maximum intensity projections obtained by confocal microscopy (LSM800, Zeiss). The number of sperm bound per mouse oocyte/embryo or per 20-µm² area (mouse and zebrafish oocytes/embryos) was obtained using the ZEN 3.2 software (Blue Edition, Zeiss). The micropyle region was defined as the area within a 160 µm radial distance from the micropyle opening, based on measurements obtained through confocal microscopy using ZEN Lite software (Zeiss, Germany). Sperm located within this defined region were classified as being in the micropyle area, while those beyond it were designated as 'away' from the micropyle. Descriptive statistics were used for comparisons of mouse sperm bound to oocytes, embryos, or agarose beads and for mouse sperm that have crossed the zonae with One-Way ANOVA followed by Tukey HSD (honestly significant difference) post-hoc test analyses. The number of sperm was represented as bars or boxplots reflecting the median (horizontal line) and data points within the 10th and 90th percentiles (error bars) for at least three independent experiments. Boxes included the middle two quartiles, and outliers, when present, were indicated by dots. All statistical analyses were performed using R Studio (v. 3.6.3).

Supporting Information

Sperm Behavior in Response to Zebrafish Eggs: Observations Across Four Videos

This study investigates the behavior of mouse sperm interacting with zebrafish eggs using live imaging techniques, with a specific focus on the region surrounding the micropyle. To capture sperm-egg interactions, four separate video recordings were taken. The recordings spanned a total period of 63 minutes and 17 seconds, with each video lasting an average of 16 ± 2 minutes. In total, 546 sperm were observed across the four videos. The results showed that only a subpopulation of sperm was able to enter the micropyle canal (8.25 ± 3.63, 5.92 % ± 2.38 %), while the majority were

found not entering the micropyle and either swimming close to the micropyle without entering (110.5 ± 44.04 , $79.20\% \pm 5.42\%$) or passing by and completely ignoring the zebrafish oocyte (17.75 ± 4.22 , $15.00\% \pm 6.28\%$)(quantification performed as displayed in Suppl Figure 3).

Zebrafish sperm cannot bind mouse zonae pellucidae

To evaluate whether zebrafish sperm can recognize and bind to the mammalian zona pellucida, we performed cross-species insemination experiments using ovulated, cumulus-free mouse oocytes incubated in water with zebrafish sperm. Ovulated mouse MII oocytes ($n = 10$; 3 replicates) were denuded of cumulus cells using hyaluronidase, then gently acclimated from standard mouse IVF conditions (HTF/HSA) to Hank's solution and subsequently to water, to mimic the zebrafish insemination environment. Then, mouse eggs were incubated with zebrafish sperm using fish IVF conditions as described in the Methods section. No zebrafish sperm bound to the mouse zona pellucida (Suppl. Figure 2). These results support the hypothesis that zebrafish sperm do not interact with mouse zonae.

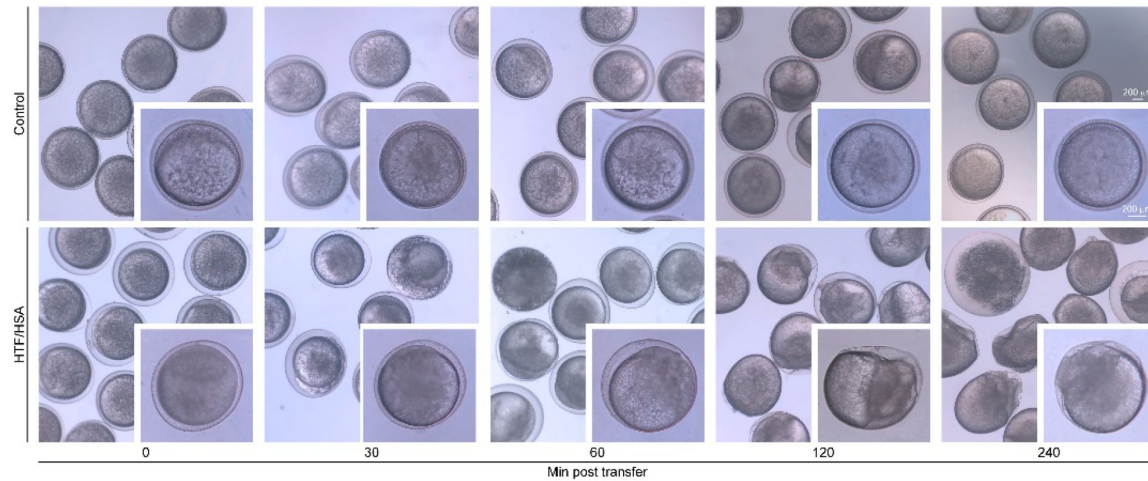
AlphaFold2-multimer modeling of cross-species interactions mediating gamete membrane adhesion

To predict the protein-interactions we used 'AlphaFold2 mmseqs2' from ColabFold (<https://github.com/sokrypton/ColabFold>). The amino sequence of each protein was obtained from the UniProt data using the following IDs: mouse Spaca6 (E9Q8Q8), mouse Izumo1 (Q9D9J7), mouse Tmem81 (Q9D5K1), mouse Izumo1R (Q9EQF4), zebrafish Bouncer (P0DPQ9), zebrafish Izumo1 (A0A2R8QH6), zebrafish Spaca6 (A0A8M9PDM1), zebrafish Tmem81 (B8JI67). AlphaFold2 mmseqs2 was run using parameters num_relax=1, num_recycles=3. The Predicted aligned error (PAE) heatmaps and interface-predicted template modeling (ipTM) scores were used for evaluation. The distribution of the ipTM scores of the top 3 structural models between the different complexes were compared using two-sided t-test. Based on AlphaFold2-Multimer structural predictions, we observed low binding affinity between zebrafish Bouncer and the mouse trimer (Izumo1 + Spaca6 + Tmem81) as indicated by low ipTM scores and high PAE values (Suppl. Figure 7).

Supplementary Figure 1.

Zebrafish eggs in HTF/HSA.

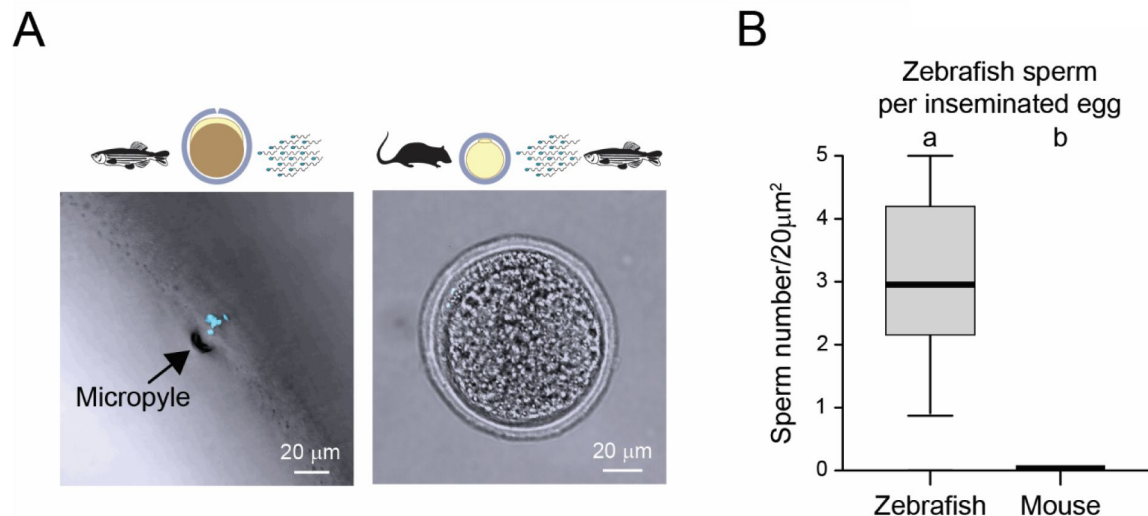
A) Zebrafish eggs maintained either in Hank's solution at room temperature (top) or under mouse IVF conditions (bottom) for 240 min

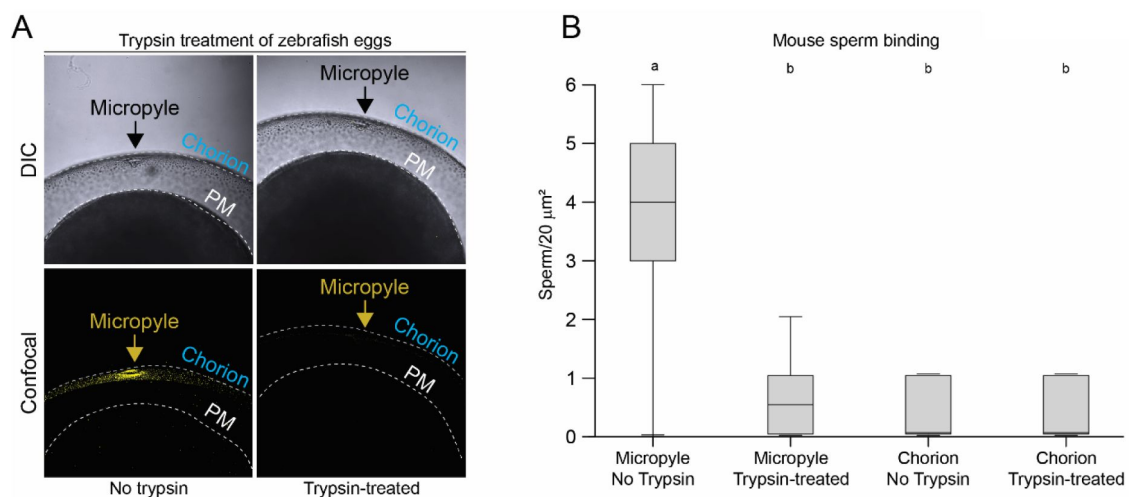


Supplementary Figure 2.

Cross species insemination of ovulated mouse MII oocytes with zebrafish sperm.

A) Left, Hoechst-stained zebrafish sperm (light blue) in the micropyle of freshly ovulated oocytes. Right, mouse MII oocytes inseminated with Hoechst-stained zebrafish sperm. **B)** Quantification of the observations shown in A, presented as boxplots illustrating the median number of mouse sperm bound to mammalian/fish peptide-coated beads or control beads. Error bars represent the 10th and 90th percentiles, while the boxes indicate the interquartile range (25th to 75th percentiles). Superscript letters indicate statistically significant differences ($P < 0.05$) as determined by one-way ANOVA followed by Tukey's HSD post hoc test.

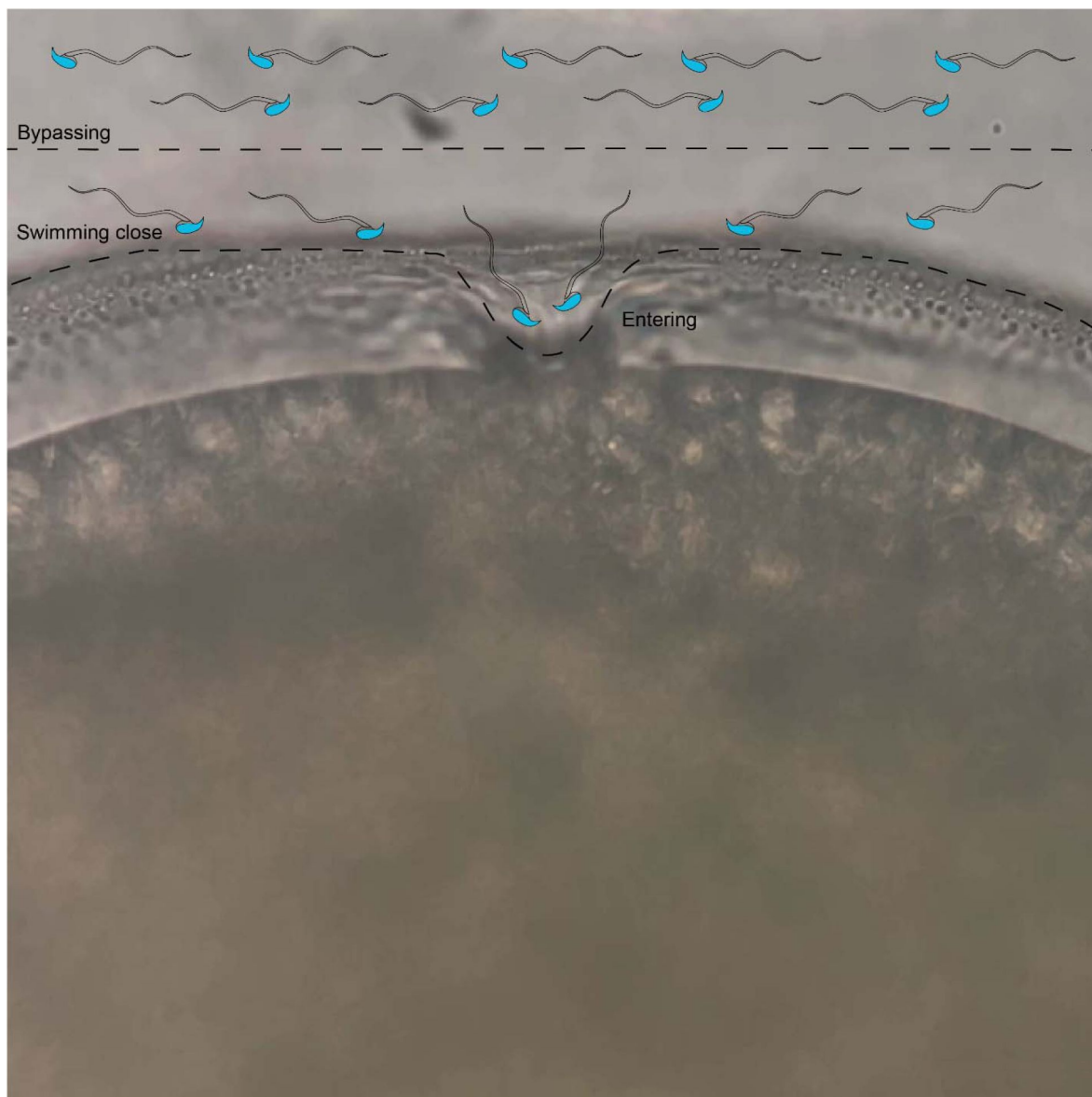




Supplementary Figure 3.

Trypsin treatment of zebrafish micropyle.

A) Zebrafish eggs untreated (left) or treated with trypsin to eliminate the MP (right); top panels, DIC images, bottom panel, confocal images. Yellow is the MP stained with WGA-633. **B)** Same as in 1G, with zebrafish trypsin-treated vs. untreated.



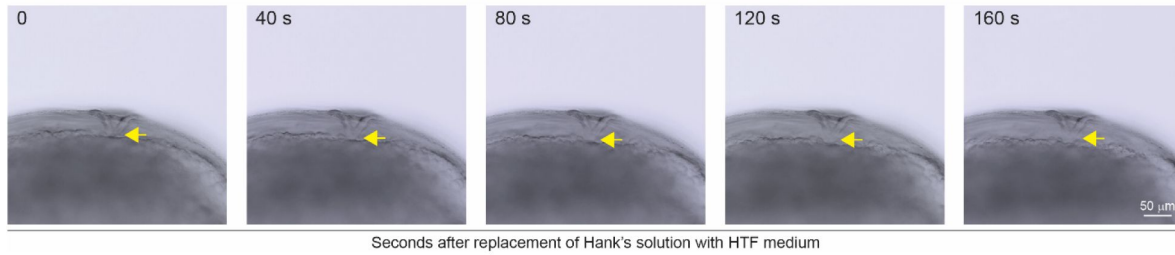
Supplementary Figure 4.

Mouse sperm bypassing the oocyte, swimming close without interacting with the micropyle or entering the micropyle canal.

Supplementary Figure 5.

Time-lapse confocal imaging of micropyle detachment under mouse IVF conditions.

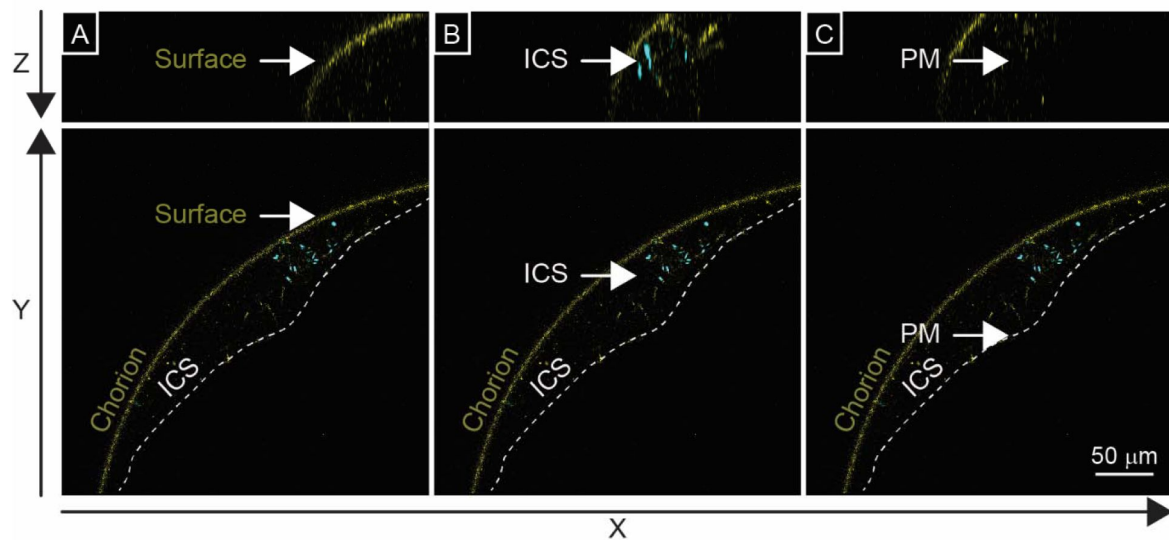
Arrows indicate progressive separation of the micropyle from the egg cell. Seconds, "s".

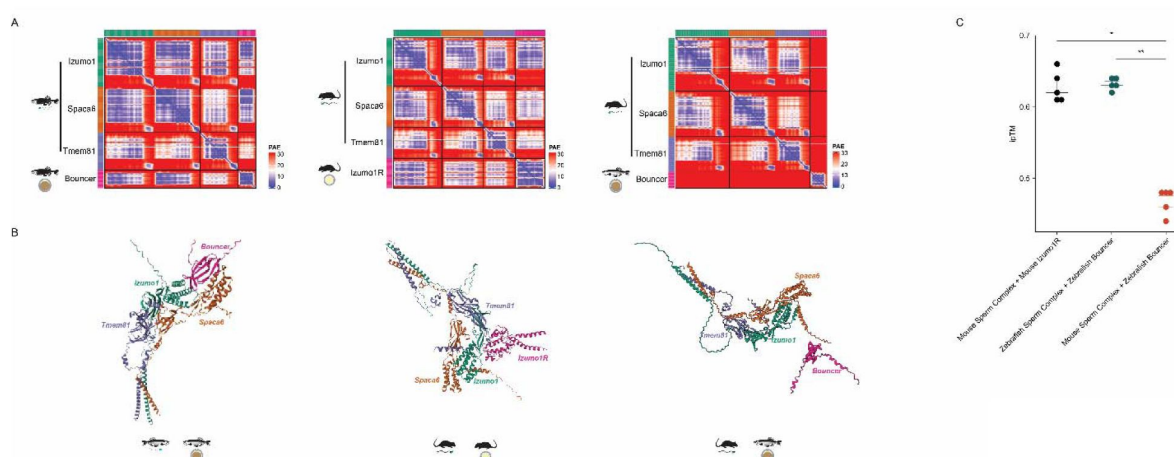


Supplementary Figure 6.

Orthogonal view generated using ZEN Lite software (Zeiss, Germany), showing z-stack sections at three focal planes.

Axes: bottom panels = XY, top panel = XZ. (A) Surface of the WGA-stained chorion; (B) Mid-region of the ICS; (C) Oocyte plasma membrane (PM). Sperm nuclei stained with Hoechst appear in light blue.





Supplementary Figure 7.

Interaction of conserved mouse and fish sperm trimmers with mammalian or fish egg proteins.

(A) Heatmaps showing the Predicted Aligned Error (PAE) plots for the top scoring model in AlphaFold2-Multimer of the interaction between the fish or mouse trimmers (Izumo1 + Spaca6 + Tmem81) and the zebrafish Bouncer or mouse Izumo1R. Blue indicates low, Red high PAE scores. **(B)** Predicted 3D structure of the interactions in (A). **(C)** Box plots showing the distribution of interface-predicted template modeling (ipTM) scores. Asterisks denote statistically significant differences ($p < 0.001$) as determined by a two-sided t-test.

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Additional files

Supplementary Video 1. *The entry of mouse sperm into the micropyle of an ovulated zebrafish egg. This time-lapse video captures the moment the first sperm enters the micropyle of a freshly ovulated zebrafish egg. At the 1-second mark, two sperm are visible in the field of view. One sperm displays an abnormal structure, with a partially coiled tail, while the other sperm is attracted to the micropyle. Within 6 seconds, the structurally normal sperm successfully navigates to and enters the micropyle canal.* [🔗](#)

Supplementary Video 2. *Simultaneous entry of two sperm into the micropyle canal of a zebrafish egg. Simultaneous entry of two sperm into the micropyle of a freshly ovulated zebrafish egg. Initially, both sperm interact with the wall of the micropyle. After 12 seconds, they move together into the micropyle canal opening.* [🔗](#)

Supplementary Video 3. *Mouse sperm interaction with zebrafish egg's inter-chorion space (ICS). Motility pattern of mouse sperm within the zebrafish egg's ICS (10x magnification).* [🔗](#)

Supplementary Video 4. *Mouse sperm in the zebrafish ICS. Same as in Supplementary video 3, 20 x magnification.* [🔗](#)

Supplementary Video 5. *High-resolution view of mouse sperm zebrafish ICS area. Same as in Supplementary video 3, 40 x magnification.* [🔗](#)

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Reviewer #1 (Public review):

Summary:

The paper is well written and investigates the cross-species insemination of fish eggs with mouse sperm. and I have a few major and minor comments.

Strengths:

The experiments are well executed and could provide valuable insights into the complex mechanisms of fertilization in both species. I found the information presented to be very interesting,

Weaknesses:

The rationale of some of the experiments, in particular those using CatSper KO sperm is, in my view.

<https://doi.org/10.7554/eLife.106303.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Summary:

The paper is well written and investigates the cross-species insemination of fish eggs with mouse sperm. I have a few major and minor comments.

Strengths:

The experiments are well executed and could provide valuable insights into the complex mechanisms of fertilization in both species. I found the information presented to be very interesting,

Thank you.

Weaknesses:

The rationale of some of the experiments is not well defined.

Thank you. In the revised manuscript, we have clarified and expanded the rationale behind each experiment to better highlight the specific questions being addressed and how each approach contributes to our overall investigation. These clarifications have been integrated throughout the Results and Discussion sections. We provide detailed rationale in our point-by-point responses to both reviewers, outlining how each experimental design was motivated by prior findings, hypotheses, or specific gaps in knowledge. We hope these revisions make the experimental logic and progression better defined and more compelling.

Major Comments:

(1) Figure 5

I do not understand the rationale for performing experiments using CatSper-null sperm and CD9-null oocytes. It is well established that CatSper-null sperm are unable to penetrate the zona pellucida (ZP), so the relevance of this approach is unclear.

We thank the reviewer for this comment. This experiment was conducted as the basis to then evaluate the contributions of progressive and hyperactivated motility to the ability of mouse sperm to locate and traverse the zebrafish micropyle. In earlier experiments (Figures 1 and 3), we assessed whether sperm-micropyle interaction was robust by comparing it to binding to the mouse zona pellucida and testing whether both interactions persisted after washing, which is standard approach to distinguish specific binding from non-specific adherence (Avella et al., 2014; Baibakov et al., 2012). Thus, we extended this analysis to CatSper1^{Null} sperm; CatSper1^{Null} sperm were still capable of binding the zona pellucida comparably to heterozygous controls, though they were unable to cross the zona of Cd9^{Null} eggs. These observations served as a validation step for the use of CatSper1^{Null} sperm for downstream micropyle interaction assays. Thus, we proceeded to test whether hyperactivated motility, absent in CatSper1^{Null} sperm, is required for locating and crossing the micropyle.

It is indeed well established that CatSper1^{Null} sperm are unable to penetrate the zona pellucida, and previous studies have typically used the absence of fertilized eggs as a readout. However, failed fertilization may result from multiple factors, including impaired sperm motility, reduced capacity to bind the zona pellucida, or an inability to penetrate it. To our knowledge, no study has quantitatively assessed the number of CatSper-deficient sperm that successfully bind, cross the zona and reach the perivitelline space. To address this, we first used normal oocytes for sperm binding and Cd9^{Null} oocytes (Le Naour et al., 2000), which allow direct quantification of sperm accumulation in the perivitelline space. We have included a detailed explanation in the Results to clarify this point, lines 352-365 and 376-369.

(2) Micropyle penetration and sperm motility

CatSper-null sperm are reportedly unable to cross the micropyle, but this could be due to their reduced motility rather than a lack of hyperactivation per se. Were these experiments conducted using capacitated or non-capacitated spermatozoa? What was the observed motility of CatSper-null sperm during these assays? Clarifying these conditions is essential to avoid drawing incorrect conclusions from the results.

Thank you for raising these points. Under our IVF conditions, qualitative observations confirmed that CatSper1^{Null} sperm displayed progressive motility, maintained sufficient

progressive motility during the first hour post-insemination and exhibited zona binding efficiency comparable to that of CatSper1^{Het} controls (Figure 5A and B). This is consistent with previous reports showing that within the first 90 minutes of sperm incubation in media, approximately 20% of CatSper1^{Null} sperm preserve motility (Qi et al., 2007). Given previous studies indicating that 15–35% of sperm undergo hyperactivation within 90 minutes (Goodson et al., 2011), and considering that 100,000 progressively motile sperm were used for insemination, we estimate that approximately 3,000 hyperactivated CatSper1^{Null} sperm were present in the cross-species insemination dish (mouse sperm x zebrafish eggs). Based on these numbers, we would have expected at least some sperm to locate the micropyle if hyperactivation were not required for its detection and entry. Nevertheless, CatSper1^{Null} sperm were detected in proximity to the micropyle canal, its opening, or within the inter-chorion space (ICS). These observations support the conclusion that the inability of CatSper1^{Null} sperm to locate and enter the micropyle is attributable to their failure to hyperactivate. Also, all sperm used in these assays were exposed to identical capacitating conditions (HTF/HSA, 37 °C, 5% CO₂). We now clarify this in the Methods, line 624, and we added more rationale under the Results, lines 361-365 and in the Discussion, lines 470-483.

(3) Rheotaxis and micropyle navigation

Previous studies have shown that CatSper-null sperm fail to undergo rheotaxis. Could this defect be related to their inability to locate and penetrate the micropyle? Exploring a potential shared mechanism could be informative.

Thank you for raising this interesting point. Indeed, homozygous mutant mice lacking expression of a different component of the CatSper channel, CatSperz, show reduced rheotactic efficiency and severe subfertility (Chung et al., 2017). We cannot exclude that complete lack of CatSper as shown in CatSper1^{Null} mice could lead to reduced rheotactic efficiency, hence we include this interpretation in the Discussion (lines 484-486).

(4) Lines 61-74

This paragraph omits important information regarding acrosomal exocytosis, which occurs prior to sperm-egg fusion. Including this detail would strengthen the discussion.

Thank you. We have revised the text in the discussion to describe the process of acrosome exocytosis, and its relevance for fertilization (lines 504-518).

Reviewer #2 (Public review):

Summary:

Garibova et al. investigated the conservation of sperm recognition and interaction with the egg envelope in two groups of distantly related animals: mammals (mouse) and fish (zebrafish). Previous work and key physiological differences between these two animal groups strongly suggest that mouse sperm would be incapable of interaction with the zebrafish egg envelope (chorion) and its constituent proteins, though homologous to the mammalian zona pellucida (ZP). Indeed, the authors showed that mouse sperm do not bind recombinant zebrafish ZP proteins nor the intact chorion. Surprisingly, however, mouse sperm are able to locate and bind to the zebrafish micropyle, a specialized canal within the chorion that serves as the egg's entry point for sperm. This study suggests that sperm attraction to the egg might be highly conserved from fish to mammals and depends on the presence of a still unknown glycosylated protein within the micropyle. The authors further demonstrate that mouse sperm are able to enter the micropyle and accumulate within the intrachorionic space, potentially through a CatSper-dependent mechanism.

Strengths:

The authors convincingly demonstrate that mouse sperm do not bind zebrafish ZP proteins or the chorion. Furthermore, they make the interesting observation that mouse sperm are able to locate and enter the zebrafish micropyle in an MP-dependent manner, which is quite unexpected given the large evolutionary distance between these species, the many physiological differences between mouse and zebrafish gametes, and the largely different modes of both fertilization and reproduction in these species. This may indicate that the sperm chemoattractant in the egg is conserved between mammals and fish; however, whether zebrafish sperm are attracted to mouse eggs was not tested.

Thank you. We performed an additional experiment with fish sperm used to inseminate ovulated mouse eggs, and results are reported in lines 183-187 and in Supplementary Figure 2.

Weaknesses:

The key weakness of this study lies in the rationale behind the overall investigation. In mammals, the zona pellucida (ZP) has been implicated in binding sperm in a taxon-specific manner, such that human sperm are incapable of binding the mouse ZP. Indeed, work by the corresponding author showed that this specificity is mediated by the N-terminal region of the ZP protein ZP2 (Avella et al., 2014). The N-termini of human and mouse ZP2 share 48% identity, which is higher than the overall identity between mouse and zebrafish ZP2, with the latter ortholog entirely lacking the N-terminal domain that is essential for sperm binding to the ZP. Given this known specificity for mouse vs. human sperm-ZP binding, it does not follow that mouse sperm would bind ZP proteins from not only a species that is much more distantly related, but also one that is not even a mammal, the zebrafish. Furthermore, the fish chorion does not play a role in sperm binding at all, while the mammalian ZP can bind sperm at any location. On the contrary, the zebrafish chorion prevents polyspermy by limiting sperm entry to the single micropyle.

We thank the reviewer for this detailed comment. In this study, our goal was precisely that one of validating the hypothesis that mouse sperm would not bind either recombinant fish ZP proteins or the chorion; in addition, we found it important to examine the observation that mouse sperm could detect the micropyle. We further elaborated this rationale in the Introduction (lines 93-100).

In addition, though able to provide some information regarding the broad conservation of sperm-egg interaction mechanisms, the biological relevance of these findings is difficult to describe. Fish and mammals are not only two very distinct and distantly related animal groups but also employ opposite modes of fertilization and reproduction (external vs. internal, oviparous vs viviparous). Fish gametes interact in a very different environment compared to mammals and lack many typically mammalian features of fertilization (e.g., sperm capacitation, presence of an acrosome, interaction with the female reproductive tract), making it difficult to make any physiologically relevant claims from this study. While this study may indicate conserved mechanisms of sperm attraction to the egg, the identity of the molecular players involved is not investigated. With this knowledge, the reader is forced to question the motivation behind much of the study.

We thank the reviewer for their perspective, and we appreciate the opportunity to further elaborate on our rationale. As outlined in our Results and Discussion sections, a growing body of evidence supports the presence of conserved molecular players and signaling pathways involved in gamete interaction across species with diverse reproductive strategies. While

zebrafish and mice do differ in their fertilization environments and modes of reproduction, these differences may not necessarily exclude the possibility of conserved molecular mechanisms underlying gamete interaction. For example, the CatSper calcium channel, which plays a key role in regulating sperm motility and hyperactivation, is conserved across a broad range of taxa—from echinoderms such as sea urchins (external fertilizers)(Seifert et al., 2015) to mammals, including mice and humans (internal fertilizers)(Lishko and Mannowetz, 2018). Moreover, sperm from some fish species possess acrosomes that undergo exocytosis prior to fertilization while sperm cross the micropyle (Psenicka et al., 2010). Also, in ovoviviparous species with internal fertilization, such as the black rockfish, sperm do undergo molecular changes while in the female reproductive tract—including immunomodulatory adaptations, glycocalyx remodeling, and interactions with ovarian cells—enabling the sperm with a longer-term survival and a selective persistence that ensures only the fittest sperm can successfully fertilize eggs (Li et al., 2024). As per the mammalian capacitation, it is broadly defined as the process during which sperm undergo hyperactivation (Yanagimachi, 1970), and acquire the ability to undergo the acrosome exocytosis, making the sperm competent for gamete fusion and fertilization (Bhakta et al., 2019; Puga Molina et al., 2018; Yanagimachi, 1957; Yanagimachi et al., 2017). Of note, acrosome exocytosis or changes in sperm motility are not exclusive to internal fertilizers. For example, as we cite in our manuscript (and as just stated above), acrosome exocytosis has been described to occur as sturgeon sperm cross the micropyle (Psenicka et al., 2010). As per changes in flagellar motility, investigations in the Pacific herring (*Clupea* sp.) demonstrated that sperm remain nearly immotile upon release into seawater and only initiate motility when approaching the micropyle region of the egg (Yanagimachi, 1957; Yanagimachi et al., 2017). In other fish, including bitterling and zebrafish, further enhancement in sperm motility is observed as sperm approach the micropyle area (Suzuki, 1958; Yanagimachi et al., 2017). These studies suggest that functional equivalents of capacitation may exist across taxa.

We interpret the observation that mouse sperm can locate and enter the micropyle as suggesting that underlying guidance mechanisms may be more broadly conserved across distant species than previously recognized. We have now elaborated on these points in the revised Discussion (lines 531-552), and we hope the motivation behind our study is now more clearly articulated.

During fertilization in fish, the sperm enters the micropyle and subsequently, the egg, as it is simultaneously activated by exposure to water. During egg activation, the chorion lifts as it separates from the egg and fills with water. This mechanism prevents supernumerary sperm from entering the egg after the successfully fertilizing sperm has bound and fused. In this study, the authors show that mouse sperm enter the micropyle and accumulate in the intrachorionic space. Whether any sperm successfully entered the egg is not addressed, and the status of egg activation is not reported.

We appreciate the reviewer's detailed comments and the opportunity to elaborate on this important aspect for our cross-insemination assay. We interpret the reviewer's reference to "sperm entering the egg" as pertaining to sperm adhesion to the oocyte plasma membrane followed by fusion with the egg cell, two separate steps regulated by different molecular players for sperm-egg plasma membrane adhesion (Bianchi et al., 2014; Fujihara et al., 2021; Herberg et al., 2018; Inoue et al., 2005) and for fusion. It is important to note that proteins mediating gamete fusion are still unidentified in fish and mammals (Bianchi and Wright, 2020; Deneke and Pauli, 2021).

In our cross-species insemination experiments, zebrafish oocytes were maintained in Hank's solution to limit spontaneous activation; however, as the reviewer correctly notes, activation likely occurred upon exposure to HTF. While this model does not recapitulate full fertilization events, it serves as a platform to explore whether mammalian sperm can detect (within the

scope of our study) and respond (future studies) to putative evolutionarily conserved signals, such as those guiding fish sperm toward the micropyle.

While investigating cross-species sperm–oocyte fusion was not within the scope of this study and would require a distinct set of experimental approaches, we believe this question is an important one. However, we do not expect our platform to be informative for evaluating sperm adhesion to the fish oolemma or for enabling cross-species gamete fusion. In our assays focused on sperm–micropyle interaction, Hoechst staining of nuclei of transgenically-tagged acrosome sperm revealed no evidence of sperm adhesion to or fusion with the fish egg membrane (Figure 4D). Also, molecular incompatibilities may further prevent this interaction: in zebrafish, the Ly6/uPAR family protein Bouncer is expressed exclusively in the egg and is necessary for sperm–egg membrane adhesion (Herberg et al., 2018). Recent studies in zebrafish and mice have shown that a conserved trimeric complex composed of Izumo1, Spaca6, and Tmem81 on the sperm surface is required for mediating adhesion to the oocyte membrane by interacting with the mammalian oocyte receptor Izumo1R (also known as JUNO) or the zebrafish oocyte receptor Bouncer (Deneke et al., 2024). One would hypothesize that for mouse sperm to adhere to the zebrafish egg membrane, the mouse Izumo1–Spaca6–Tmem81 complex would need to establish binding with Bouncer. To explore this possibility, we performed AlphaFold2-Multimer structural predictions and docking analyses to mimic an interaction between mouse Izumo1–Spaca6–Tmem81 and zebrafish Bouncer, using mouse Izumo1–Spaca6–Tmem81 and Juno or zebrafish Izumo1–Spaca6–Tmem81 and Bouncer as positive controls. We observed low binding affinity between zebrafish Bouncer and the mouse trimeric complex (Izumo1, Spaca6, and Tmem81), as indicated by low ipTM scores and high predicted aligned error (PAE) values. These findings suggest that the mouse complex is unlikely to form an interaction with Bouncer (now shown in Suppl. Figure 7). These predictions were consistent with our observations that no sperm were found adhering or fusing to the egg cell. We describe methods and results in the supplementary files (Supporting Info, lines 53–66) and in the result sections (lines 335–339).

In Supplementary Videos 3–4, the egg shown has been activated for some time, as evident by the separation of yolk and cytoplasm, yet the chorion is only partially expanded (likely due to mouse IVF conditions). How multiple sperm were able to enter the micropyle but presumably not the egg is not addressed, yet this suggests that the zebrafish mechanism of blocking polyspermy (fertilization by multiple sperm) is not effective for mouse sperm or is rendered ineffective due to mouse IVF conditions. The authors do not discuss these observations in the context of either species' physiological process of fertilization, highlighting the lack of biological context in interpreting the results.

Thank you for raising this important point. One model for mammalian gamete recognition at the zona supports the notion that mouse sperm can penetrate extracellular matrices as long as sperm can bind to them, and binding is dependent on the cleavage status of ZP2. Zonae surrounding unfertilized mouse eggs present uncleaved ZP2 and these zonae support sperm binding. After gamete fusion, the cortical granules release ovastacin which cleaves ZP2 at the N-terminus, and consequently, zonae presenting cleaved ZP2 no longer support sperm binding. This mechanism acts as block to zona binding and prevents further crossing (Bhakta et al., 2019). Indeed, fertilized mouse eggs or 2-cell embryos surrounded by a zona containing uncleaved ZP2 support de novo sperm binding, and supernumerary sperm cross the zona and accumulate in the perivitelline space, unable to fuse with the fertilized oocyte plasma membrane or blastomere cells (Baibakov et al., 2012, 2007; Burkart et al., 2012; Gahlay et al., 2010). Thus, because under our experimental conditions, mouse sperm could interact with the micropyle opening, we interpret these findings to suggest that once interaction occurs at the micropyle opening, mouse sperm are capable of crossing it, even under conditions where the micropyle may be detached from the oocyte due to oocyte activation. Therefore, our data

indicates that mouse sperm may be able to bypass the mechanism of zebrafish oocytes blocking multiple sperm to pass through the micropyle, even after oocyte activation. This point has now been incorporated into the revised Discussion (lines 425-441).

The authors further show that the zebrafish micropyle does not trigger the acrosome reaction in mouse sperm. Whether the acrosome reacts is not correlated with a sperm's ability to cross the micropyle opening, as both acrosome-intact and acrosome-reacted sperm were observed within the intrachorionic space. While the acrosome reaction is a key event during mammalian fertilization and is required for sperm to fertilize the egg, zebrafish sperm do not contain an acrosome. Thus, these results are particularly difficult to interpret biologically, bringing into question whether this observation has biological relevance or is a byproduct of egg activation/chorion lifting that indirectly draws sperm into the chorion.

We thank the reviewer for raising this point and we appreciate the opportunity to elaborate on the biological relevance of this experiment. Our motivation to assess acrosome status in mouse sperm following entry into the zebrafish micropyle stemmed from the following biological considerations. In fish species such as the sturgeon, sperm present an acrosome and undergo acrosome exocytosis while passing through the micropyle, before gamete fusion (Alavi et al., 2012; Psenicka et al., 2010). By contrast, zebrafish sperm lack an acrosome, raising the hypothesis that the zebrafish micropyle may not be able to trigger acrosome exocytosis. However, this possibility has not been experimentally tested. We therefore considered it important to investigate whether passage through the zebrafish micropyle induces acrosome exocytosis in mouse sperm. We have revised the Discussion to better clarify the rationale behind the experiment as well as the interpretation of the findings (lines 504-518). As per the chorion lifting indirectly drawing sperm into the chorion, we have not observed this phenomenon.

The final experiments regarding CatSper1's role in mediating mouse sperm entry into the micropyle/chorion are not convincing. As no molecular interactions are described or perturbed, the reader cannot be sure whether the sperm's failure to enter is due to signaling via CatSper1 or whether the overall failure to undergo hyperactivation limits sperm motility such that the mutant sperm can no longer find and enter the zebrafish micropyle. Indeed, in Figure 5E, no CatSper1 mutant sperm are visible near any part of the egg, suggesting that overall motility is impaired, and this is not a phenotype specific to interactions with the micropyle.

We appreciate the comment and the opportunity to further elaborate on the rationale of this experiment. While our data demonstrates a lack of CatSper1^{Null} sperm accumulation within the micropyle and ICS, we appreciate that this may be interpreted as the result of general motility defects, rather than a specific failure in undergoing hyperactivation and micropyle recognition. CatSper1^{Null} sperm are known to lack hyperactivated motility and exhibit a progressive loss of forward motility over time. After 90 minutes, only ~20% of CatSper1^{Null} sperm remain motile, compared to over 70% in fertile sperm (Qi et al., 2007). Of note, under our IVF conditions, CatSper1^{Null} sperm retained sufficient progressive motility during the first hour post-insemination to bind the zona pellucida with comparable efficiency to CatSper1^{Het} controls. Based on prior reports indicating that 15–35% of sperm exhibit hyperactivation by 90 minutes (Goodson et al., 2011), and considering that we inseminated with 100,000 progressively motile sperm, we estimate that approximately 3,000 hyperactivated CatSper1^{Null} sperm were present in the dish. Yet, none were observed near the micropyle canal, its opening, or within the ICS. This led us to conclude that failure to hyperactivate underlies the inability of CatSper1^{Null} sperm to reach and traverse the micropyle. Also, we appreciate that identifying the molecular components of the micropyle would allow direct testing of whether the CatSper channel is activated in response to

micropyle-associated signals. Indeed, no targeted perturbation of molecular interaction regulating micropyle recognition was performed in this study, as the molecular identity of the zebrafish micropyle guidance cue remains unknown. Efforts to identify and characterize this factor are ongoing in our lab and lie outside the scope of the current work. Therefore, throughout the manuscript, we have clarified that it is the failure to undergo hyperactivation, rather than the absence of CatSper per se, that limits the ability of sperm to locate and traverse the micropyle. The rationale for the experiment, the interpretation of our findings, and relevant future directions have been further elaborated in the revised Abstract, Impact Statement and Discussion (lines 40-41; 46-47; 343-365; 376-379; 389-399; 470-486).

Reviewer #1 (Recommendations for the authors):

Minor Comments

(1) Figure numbering

There appear to be inconsistencies in the figure references. For example, what is referred to as Figure 3F in the text is actually Figure 4F. Please review and correct all figure labels for accuracy.

We thank the reviewer for pointing this out. We have carefully reviewed the manuscript and corrected all figure references throughout the text. Also, for better flow and coherence, we have moved the paragraph describing the videos to the end of the Results section titled "Mouse sperm recognize the micropylar region of fish oocytes." Previously, the callout of panels in Figure 3 was out of order (3A, 3B, 3E, 3C, 3D), and this reorganization also helps maintain logical progression through the figure panels.

(2) Figure 5 terminology:

The term "normal" sperm should be replaced with "CatSper heterozygous (Het)" sperm to avoid confusion and improve precision.

We thank the reviewer for this helpful suggestion. We have revised the terminology in Figure 5 and throughout the manuscript, replacing "normal" sperm with "CatSper1 heterozygous (Het)"

Reviewer #2 (Recommendations for the authors):

In addition to my comments in the public review, I would encourage the authors to consider the following suggestions:

The authors show that mouse sperm can find and enter the fish micropyle, and that this depends on the presence of MP. To better assess sperm binding to the micropyle region, the number of sperm binding to the micropyle vs. non-micropyle chorion should be clearly quantified, as well as the percentage of sperm that enter the micropyle compared to the total used for insemination. The authors state several times throughout the text that a "subpopulation" of mouse sperm finds and enters the micropyle, but it would be more precise and informative to give a percentage.

We thank the reviewer for this suggestion. We have now reported also the number of sperm bound to the other regions of the chorion (away; lines 231-233), as well as the percentage of sperm that entered the micropyle relative to the total number used for insemination (lines 276-279).

To ensure that all sperm are inside the chorion, the egg should be removed from the insemination dish, washed thoroughly, and then the chorion should be torn open to

definitively show that the sperm were indeed inside.

We thank the reviewer for these excellent suggestions. As per ensuring that the sperm are inside the ICS, (as shown now in Figures 4A, F, G, Supplementary Figure 6 and Supplementary Movies 3–5), the inseminated oocytes were thoroughly washed prior to imaging to ensure that only sperm located inside the chorion were visualized (as described in the Methods, lines 646–648). In addition, to confirm the spatial localization of sperm within the ICS, we are now including additional TEM images showing sperm in the ICS (Figure 4G, right panel). Also, we generated orthogonal views using ZEN Lite software (Zeiss, Germany) from a z-stack encompassing the full volume of the chorion, ICS, and oocyte (added in the supplementary materials, as Supplementary Figure 6). These views display three focal planes: the surface of the WGA-stained chorion, the middle of the ICS, and the oocyte plasma membrane. Sperm nuclei stained with Hoechst are clearly visible below the chorion surface and above the oocyte plasma membrane, confirming their localization within the ICS. Additionally, in a separate set of experiments, as recommended by this reviewer, we mechanically disrupted the chorion and consistently detected sperm within the ICS. This procedure, however, was technically challenging: upon disruption, the chorion often collapsed onto the oocyte, and during the extraction process, sperm were sometimes displaced. As a result, it was not always possible to determine with complete confidence whether the sperm had originally been located inside or outside the chorion. However, we hope that the additional TEM and confocal images (Figure 4G and Supplementary Figure 6) offer further support for the localization of sperm within the ICS.

I would further suggest that they examine the micropyle opening after the entry of multiple sperm, as well as the dynamics of egg activation during insemination with mouse sperm.

Thank you. We now include one additional TEM image capturing the full structure of a micropyle that was traversed by multiple mouse sperm (shown in Figure 4G, left panel).

At what point does the micropyle detach from the egg surface? Live imaging of this process with a confocal microscope would be very informative.

During live imaging, the interval between placing the oocyte in the imaging dish, replacement of Hank's solution with HTF and the addition of sperm, followed by the initiation of video acquisition, is approximately 2 to 3 min. By this time, the ICS is already apparent (Supplementary Video 2), although the micropyle appears to remain adherent to the egg cell. Partial detachment of the micropyle from the egg cell begins around 6–7 minutes after imaging starts and continues progressively over time. We provide time-lapse imaging frames to show the micropyle detachment under mouse IVF conditions (Supplementary Figure 5).

Along the same lines, sperm should be doubly labeled with an acrosome-independent marker, i.e., a live DNA stain or MitoTracker. Then the authors could track if any sperm are actually able to enter the egg itself, which would be highly unlikely but an important detail to confirm.

Thank you for pointing this out. In our assays designed to study sperm–micropyle interactions, Hoechst staining of nuclei in transgenically labeled acrosome sperm showed no indication of sperm adhesion to, or fusion with, the zebrafish egg cell (Figure 4D).

Line 242, 282: The text should refer to Figure 4, not 3. Please make sure all figure references correspond to the correct figure and panel.

Thank you for bringing this to our attention. We have carefully reviewed the manuscript and corrected the reference to Figure 4, along with all other figure and panel citations to ensure they accurately correspond to the correct content. Also, to improve the overall flow, we relocated the paragraph describing the videos to the end of the Results section titled "Mouse sperm recognize the micropylar region of fish oocytes". This change also helped correct the sequence of figure panel references, which were previously cited out of order (i.e., 3A, 3B, 3E, 3C, 3D).

Line 244: The authors quantify sperm that are "away" from the micropyle, but this is not clearly defined. This should be given as a set radius or distance from the center (e.g., in microns). If the sperm are still motile, can this be accurately measured?

We thank the reviewer for this valuable suggestion. We have now defined "away from the micropyle" as a distance greater than 160 μm from the center of the micropyle. This measurement was determined using confocal z-stack projections of fixed samples. These details have been added to the revised Methods section (lines 670-674).

To strengthen the conclusion that the sperm chemoattractant is indeed conserved from fish to mammals, the authors could show that zebrafish sperm are also able to find/approach mouse eggs. Even more compelling would be to show the same is true for other species combinations. As it stands, the choice of comparing mouse and zebrafish does not seem scientifically motivated but rather due to their availability.

We thank the reviewer for this important suggestion. To test whether zebrafish sperm are capable of binding to the mammalian zona pellucida, we conducted the suggested experiment: ovulated, cumulus-free mouse oocytes were placed in water and incubated with zebrafish sperm. We did not observe any zebrafish sperm bound to the mouse zona pellucida, consistent with the hypothesis that zebrafish sperm do not recognize or interact with mammalian zonae or ZP proteins. This has now been added in the Results (lines 183-187) and shown in Supplementary Figure 2. We interpret these findings as in cross-species insemination assays, reciprocity in sperm-egg interaction is not always observed. For example, while human sperm bind only to human zonae and not to mouse zonae, mouse sperm are able to bind both mouse and human zonae (Avella et al., 2014; Baibakov et al., 2012; Bedford, 1977). This asymmetry may reflect species-specific adaptations in sperm-egg recognition. We have now added this point to the revised Discussion to clarify the rationale and context of our approach (lines 416-423).

As per the choice of experimental models, while we agree that testing additional species combinations would broaden the scope of the findings, the choice to compare mouse and zebrafish was not solely based on availability. Rather, it was motivated by the opportunity to examine sperm guidance across two evolutionary distant vertebrates. This contrast allows us to seek for potential conservation of structural or molecular cues involved in gamete interaction. Additionally, both zebrafish and mouse offer extensive gene editing, blotting and imaging reagents, which are particularly valuable should future studies aim to identify and functionally disrupt genes encoding micropyle-associated proteins and their putative orthologs in mammals.

For the CatSper experiment, I would suggest that the authors repeat this experiment with another mouse sperm mutant that is known to have reduced/altered motility. With the current data, I do not believe the failure to find/enter the micropyle is necessarily CatSper-specific. Because we do not know what the sperm interacts with in the micropyle or what the MP interacts with on the sperm, the signaling pathway cannot be tested, making other controls necessary for these results to be meaningful.

Thank you for highlighting this important point. A wide range of mouse models with sperm motility defects exhibit subfertility or infertility due to structural abnormalities in the axoneme or midpiece rigidity. (Miyata et al., 2024). These defects often result in impaired progressive motility, failure to reach the zona pellucida, or inability to bind or penetrate it. In contrast, we could test and validate that CatSper1^{Null} sperm display preserved early progressive motility but fail to transition into hyperactivated motility, making them particularly well suited for specifically assessing the role of hyperactivation in sperm navigation toward and entry into the micropyle. Taken together, these points, along with those discussed in our response to the public review, led us to conclude that the CatSper1^{Null} model provides the most biologically relevant context currently available to assess the role of hyperactivation in guiding sperm to the micropyle.

The authors could greatly strengthen the discussion by addressing the key points I raised in the public review, particularly in terms of interpreting these results in the context of each species' physiological mode of fertilization.

We thank the reviewer for this important recommendation. We have carefully revised the Discussion to address the key points raised in the public review, particularly by framing our findings within the context of the distinct physiological modes of fertilization in each species, as indicated in our answers to the public review. We hope these additions have strengthened the manuscript as suggested.

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